Satellite Communications
ECE 4644 / 5610 Spring 2026

Week #3

Feb 2 - 6, 2026

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** Breaking News **

- Artemis II is a NASA mission to fly astronauts around the moon
- Countdown has begun for a launch as early as Feb. 8 (Sunday)
- Critical fueling test will take place this evening (Feb. 2)
- Artemis II will be the first crewed mission to move beyond LEO since 1972 (Apollo 17)

Artemis II at Launch Complex 39B (January, 2026)

Orbit Characteristics - Review

- Satellite orbits are characterized by <u>altitude</u>, often cited as LEO, MEO, or GEO
- And by the <u>inclination</u>, or tilt, of the orbit with respect to Earth's equatorial plane
- A satellite in geostationary orbit (GEO) maintains a fixed position in the sky for an observer on Earth's surface
- Orbital altitude, velocity, and period are closely related

Space@VT Seminar on VLEO

- The lowest altitude range (< 400 km) is now often labeled VLEO – Very Low Earth Orbit
- Advantages: easier to reach and operate a satellite
- Disadvantage: atmospheric drag
- Space@VT seminar tomorrow (Tuesday, Feb 3, 2-3 PM) by Dr. Scott England on the impact of atmospheric variability on VLEO satellites
- Watch for Announcement with Zoom link

Topics this Week (1/2)

- Space and Ground Segments
- Basics of: Transponders
- Frequency bands
- Forces on a satellite
- Derivation of Equation of the Orbit
- Geocentric and orbital plane coordinate systems

Topics this Week (2/2)

- Kepler's Laws
- Orbital elements
- Locating a satellite in its orbital plane
- Sky coordinates: Right Ascension and Declination
- Locating a satellite with respect to Earth
- Look angles

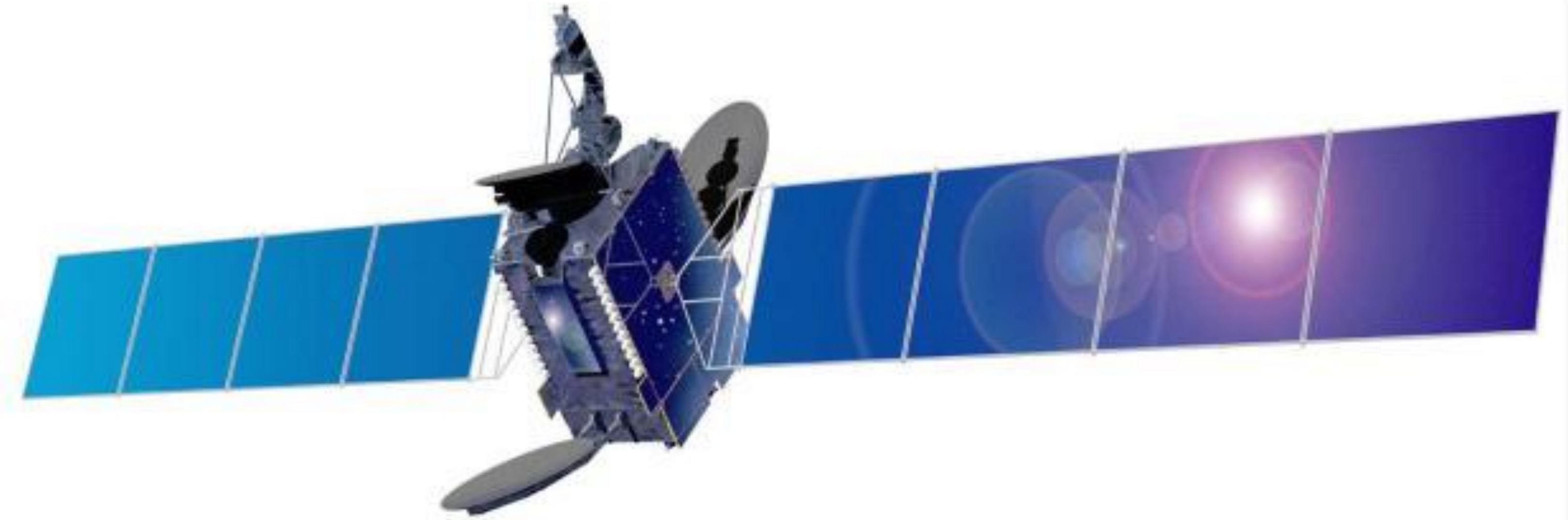
Space and Ground Segments

- *Space segment* consists of the satellite and the radio paths to and from the satellite
- *Ground segment* consists of transmitting and receiving earth stations
- A satellite communications network often has one transmitting station and many receivers
- Example:

  Direct broadcast satellite television (DBS-TV)

  - one transmit station, millions of receivers

Space segment

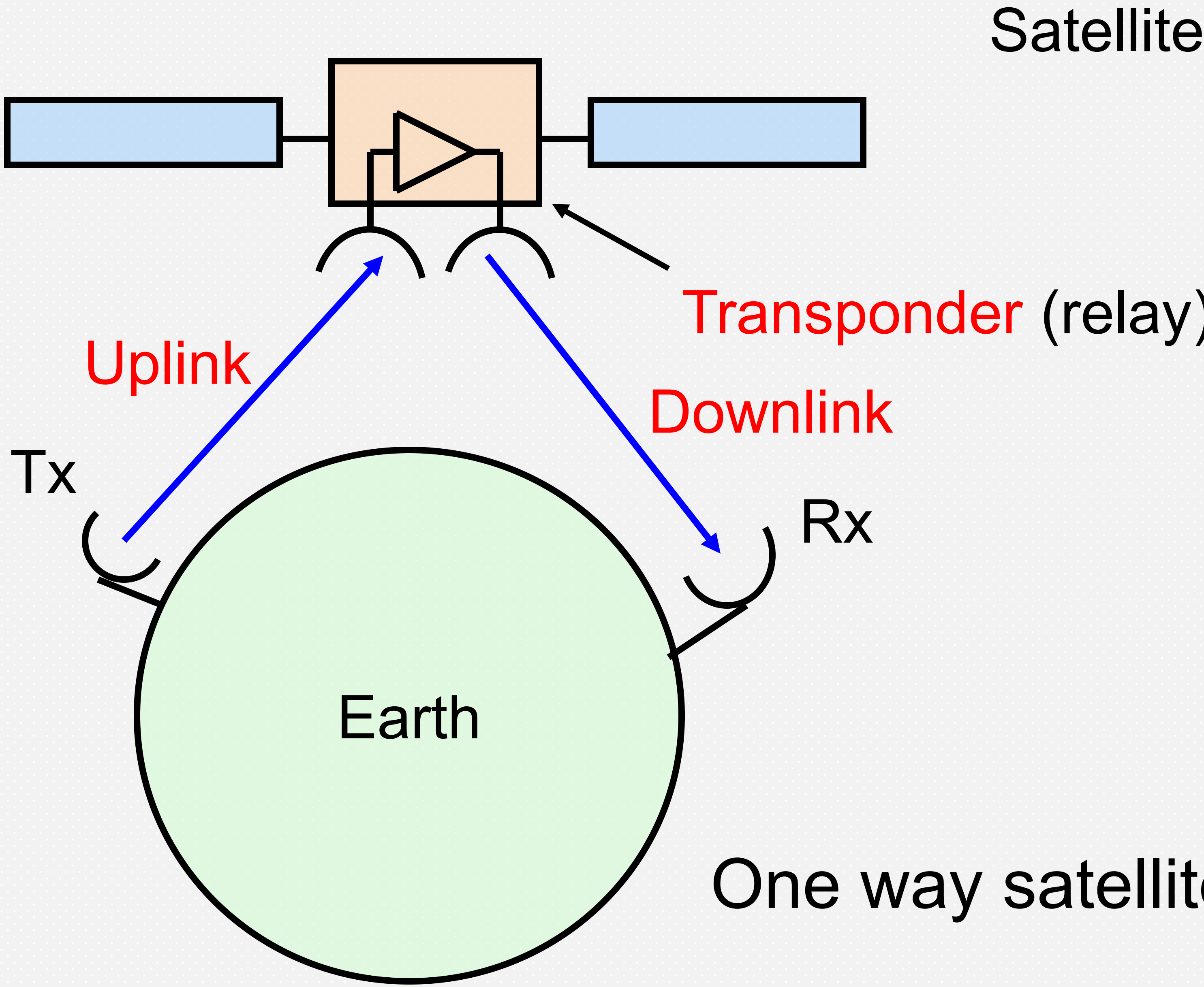


Echostar 5

A 3-axis stabilized GEO satellite

Ground segment

Echostar transmit / receive station



One way satellite link

Transponders

- Satellites typically carry many *transponders*
- Transponder relays signals back to earth
  – Signal is received, amplified in low noise amplifier (LNA)
  – Signal frequency is changed
  – Signal is amplified in high power amplifier (HPA), and then transmitted

C-Band Satellite Link

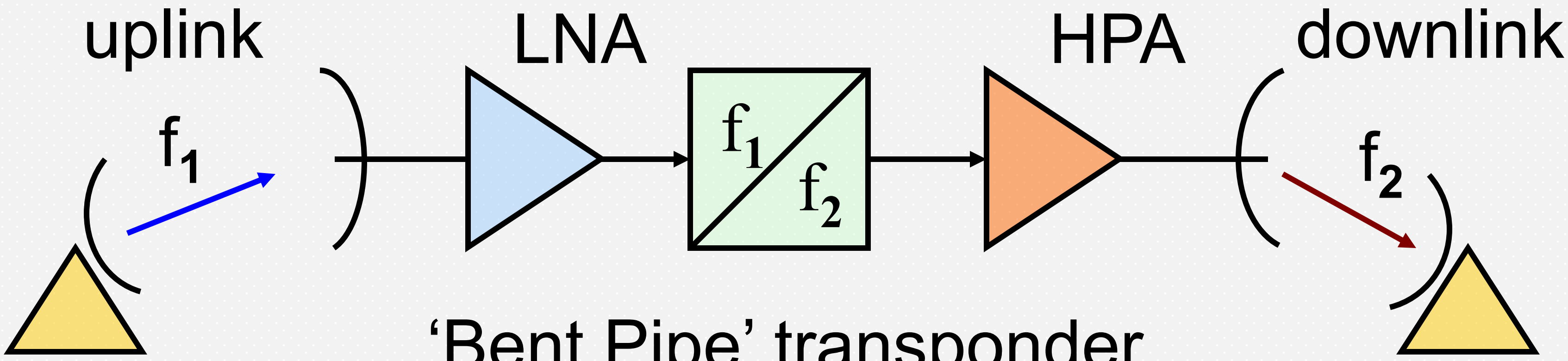
<u>Uplink signal in 6 GHz band</u>

> single conversion with LO = 2225 MHz

<u>Downlink signal in 4 GHz band</u>

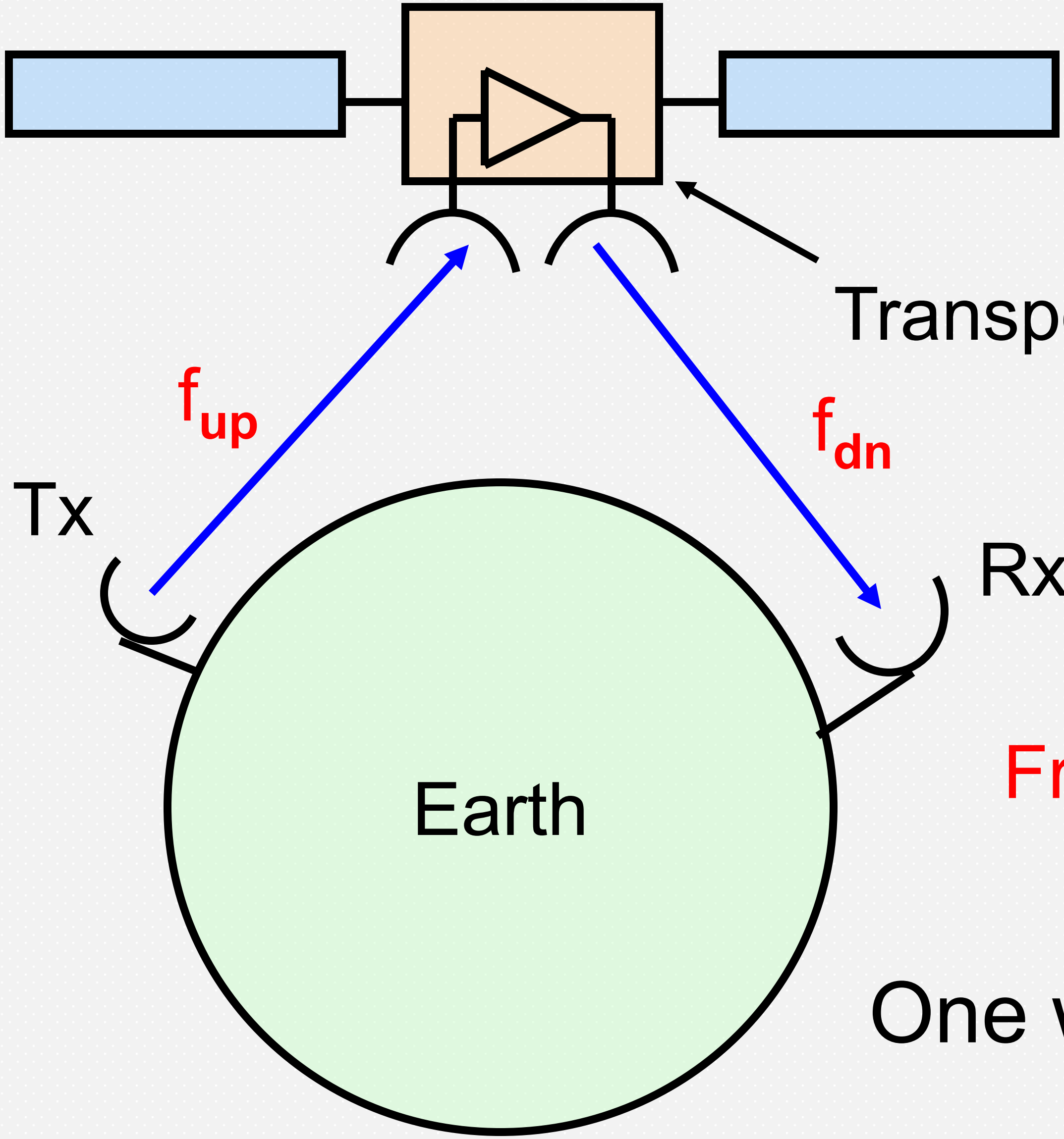
E.g. Satellite receives at f_1 = 6000 MHz

Transmits at f_2 = 6000 – 2225 = 3775 MHz



'Bent Pipe' transponder

Satellite



Transponder

Frequency band

One way satellite link

Microwave Frequency Bands

Frequency	Band	Typical Wavelength	Apps
1 – 2 GHz	L	30 cm	GPS
2 – 4 GHz	S	10 cm	ISS, XM radio
4 – 8 GHz	C	5 cm	Sat comm
8 – 12 GHz	X	3 cm	FSS, MSS
12 – 18 GHz	Ku	2 cm	Sat comm
18 – 27 GHz	K		
27 – 40 GHz	Ka	1 cm	Sat comm
40 – 75 GHz	V	mm	Sat comm

Band system was developed for radar but is widely used

Satellite Frequency Bands

- The International Telecommunications Union Radiocommunication Sector (ITU-R) allocates radio frequencies
- World Radio Conference held every few years
- The ITU is an agency of the United Nations
- National agencies manage frequency allocations and spectrum usage

Satellite Frequency Bands

- In the U.S., two governmental organizations manage radio frequency spectrum
- The Federal Communications Commision (FCC) manages non-federal use
- The National Telecommunications and Information Administration (NTIA) manages federal use

Satellite Frequency Bands

- Major satellite bands:

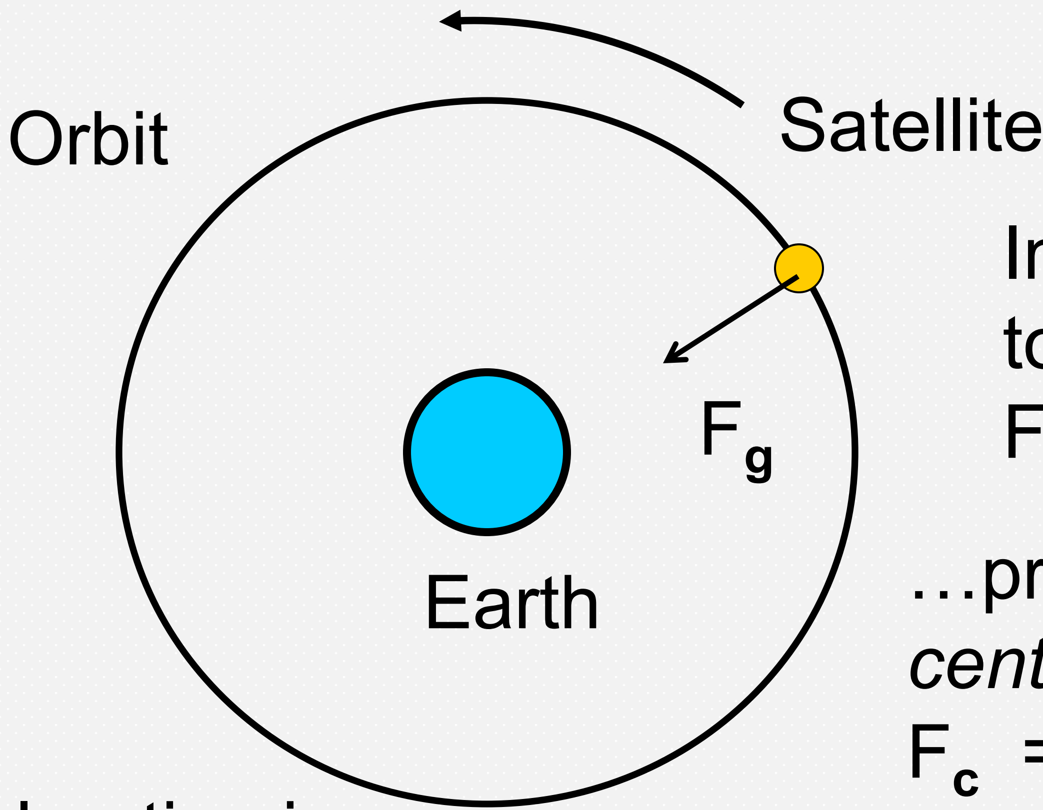
  Fixed satellite service (FSS)

  - C band: uplink 6 GHz downlink 4 GHz
  - Ku band: uplink 14 GHz downlink 11 GHz
  - Ka band: uplink 30 GHz downlink 20 GHz

Force of Gravity on a Satellite

- Force = Mass $\times$ Acceleration
- Unit of Force is a *Newton*
- A *Newton* is the force required to accelerate 1 kg by 1 m/s^2
- Underlying units of a *Newton* are therefore (kg) $\times$ (m/s^2)
- A satellite in orbit experiences an attractive force due to gravity

Force of Gravity on a Satellite



Inward force due to gravity
$F_g = GM_E m / r^2$ …

…provides the *centripetal force*
$F_c = mv^2 / r$
to maintain a stable circular orbit

Acceleration is
$a = v^2 / r$

Balance of Forces (?)

- Satellite is pulled towards center of earth
- Satellite texts often refer to an inward gravity force and an outward *centrifugal force*. The latter is not a true physical force but a device to explain the orbit in terms of 'balancing' inward and outward forces, i.e.,

$$F_g = F_{in} = F_{out} \quad \text{with} \quad F_{out} = mv^2 / r$$

- Either way, for a stable orbit we have

$$GM_E m / r^2 = mv^2 / r$$

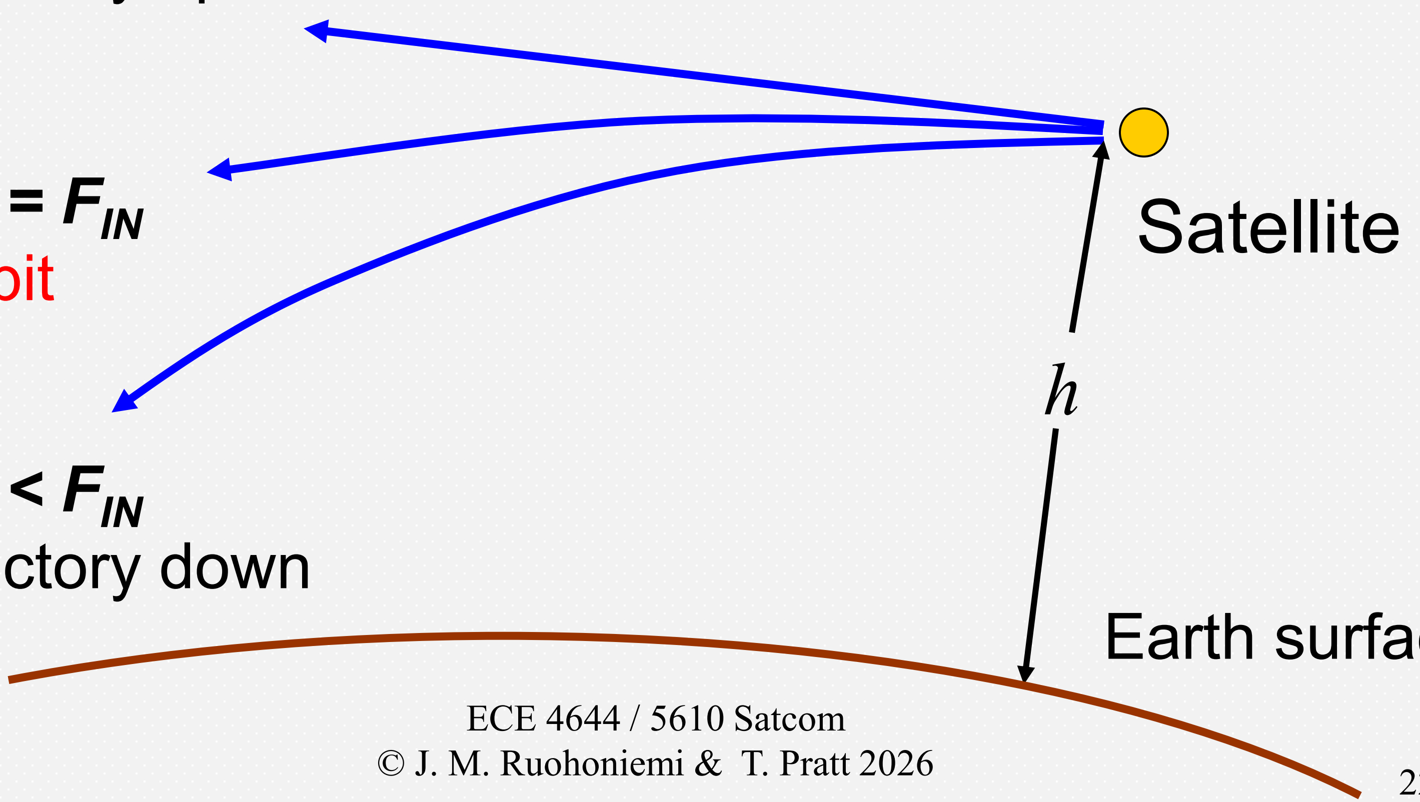
In Orbit?
Outward force versus Inward force

***F_{OUT}* depends on sat velocity**

$F_{OUT} > F_{IN}$
Trajectory up

$F_{OUT} = F_{IN}$
In orbit

$F_{OUT} < F_{IN}$
Trajectory down



Balance of Forces

- The force balance equation can be reduced to

$$GM_E / r = v^2$$

or $\quad v = (\mu/r)^{1/2}$

where $\quad \mu = GM_E$ is *Kepler's constant*

- Orbital velocity and orbital radius are connected
- The satellite orbit is set in the first place by the choice of velocity
- The orbit can be modified subsequently by changing the velocity

Acceleration Formula

- a = acceleration due to gravity = μ / r^2 km/s^2
- r = radius from center of earth
- μ is Kepler's constant = 3.9860044×10^5 km^3/s^2
- μ = universal gravitational constant G multiplied by the mass of the earth M_E
- G = 6.672×10^{-11} Nm^2 / kg^2
- Mind the distance units, are they meters [m] or kilometers [km]?

Satellite orbits

- If the orbit is circular, the period, T_p , is given by

$$T_p = 2\pi r / v = (2\pi r^{3/2}) / \mu^{1/2}$$

- The period depends only on the radius of the orbit, r, and not on the mass of the satellite

Orbit Limits

- For a stable orbit there must be a balance between inward gravitational force and outward ‘centrifugal’ force
- Must not be too close to the earth or the satellite will be slowed by the atmosphere (so, above 250 km altitude)
- Velocity must be between 18,000 mph and 25,000 mph (note departure from metric units)
- Velocity above 25,000 mph allows s/c to escape from Earth’s gravity

Orbital Velocities and Periods

Satellite System	Orbital Height (km)	Orbital Velocity (km/s)	Orbital Period hr	min	sec
GEO	35,786.43	3.0747	23	56	4.091
ICO-Global (MEO)	10,255	4.8954	5	55	48.4
Iridium (LEO)	780	7.4624	1	40	27.0

Coordinate Systems

- To solve the equation of motion we need a way to designate the position of the satellite and express its orbit
- Gravitational pull is towards the center of earth
- Orbital coordinates are defined with center of earth as origin
- Consider *geocentric* coordinate system first

Geocentric coordinate system

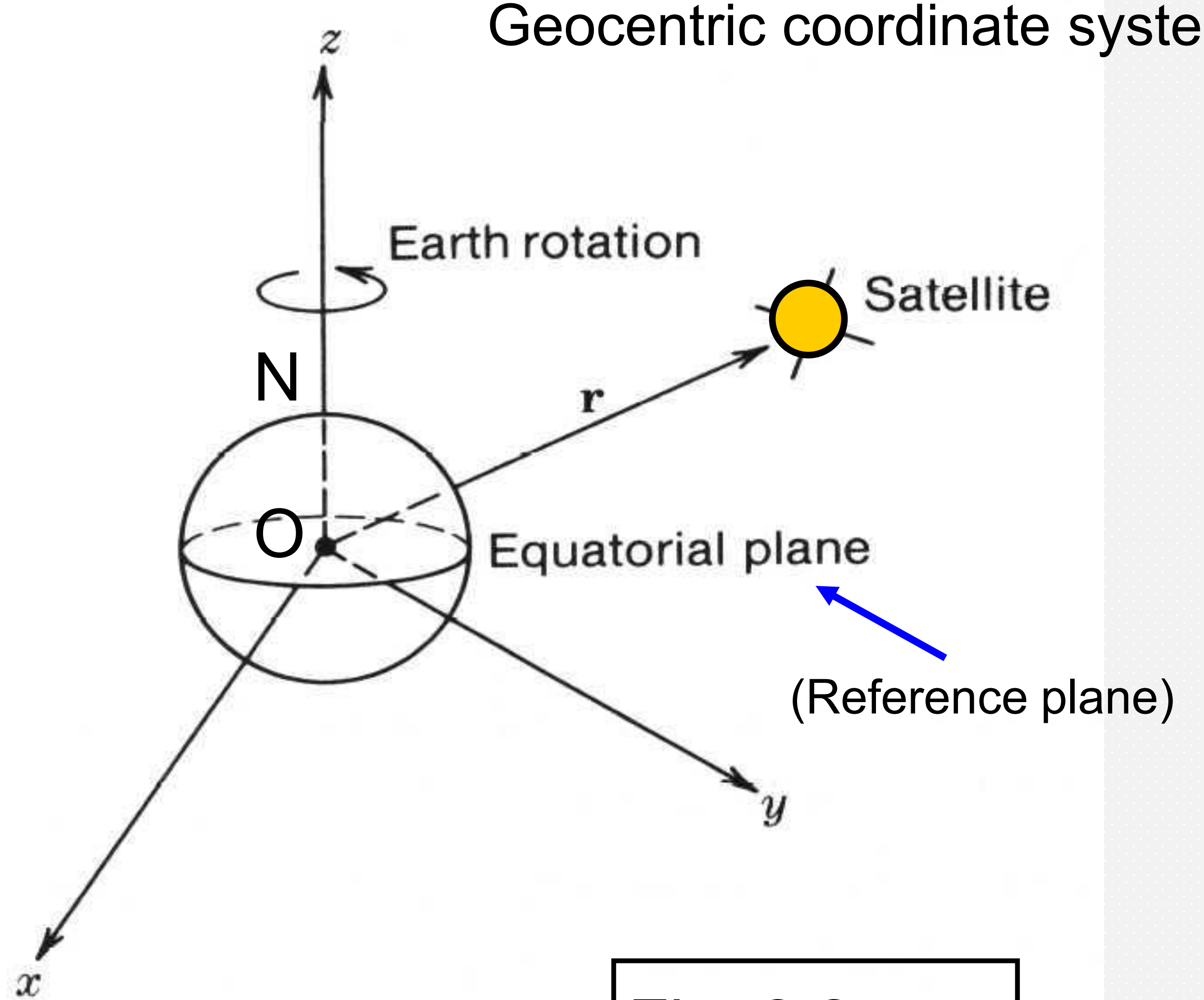


Fig. 2.2 text

Orbital Analysis

Newton's second law (**F** = m**a**) renders*

$$\mathbf{F} = -\frac{G\,M_E m}{r^2}\hat{\mathbf{r}} = m\frac{d^2\mathbf{r}}{dt^2} = m\frac{d^2 r\hat{\mathbf{r}}}{dt^2}$$

Eqns (2.7, 2.8)

With substitution we have

$$-\frac{\hat{\mathbf{r}}}{r^2}\mu = \frac{d^2 r\hat{\mathbf{r}}}{dt^2}$$

Eqn (2.9)

$$\frac{d^2 r\hat{\mathbf{r}}}{dt^2} + \frac{\hat{\mathbf{r}}}{r^2}\mu = 0$$

Eqn (2.10)

* Eqns modified slightly from text

Satellite Orbits

- Our goal is a solution for **r** in terms of *t*
- Equation 2.10 is a second order differential equation in **r**
- Solution has six constants called the *orbital elements*
- Solution is an orbit that lies in a plane and has constant angular momentum
- Leads to *Kepler's three laws of planetary motion*

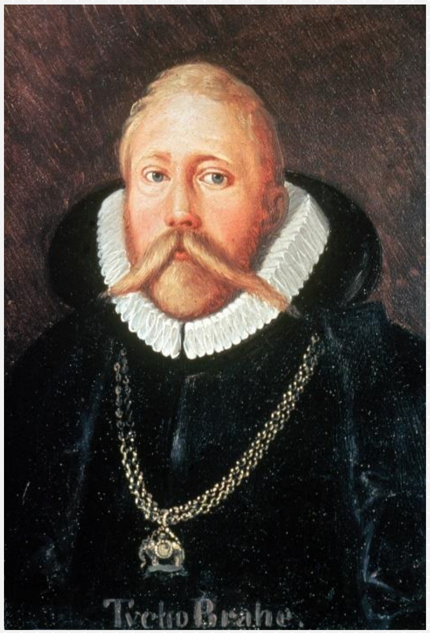
Kepler's Three Laws

- The orbit of a satellite is an ellipse with Earth at one focus
- The satellite sweeps out equal arc areas in equal times. For an ellipse, this means that the orbital velocity varies around the orbit
- The square of the period of revolution is proportional to the cube of the semi-major axis of the ellipse

Satellite Orbits

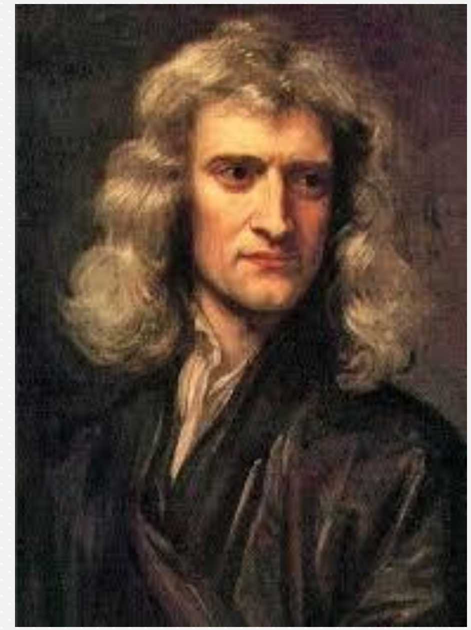
- Johannes Kepler (1571 - 1630) - German astronomer and mathematician to the Emperor of Austria
- Kepler derived his laws based on observations of planets by the Danish astronomer and nobleman Tycho Brahe (1546 - 1601)
- Theirs was a remarkable collaboration between an experimentalist and a theorist

Satellite orbits



Tycho Brahe
(1546 – 1601)

Johannes Kepler
(1573 – 1630)



Issac Newton
(1642 – 1727)

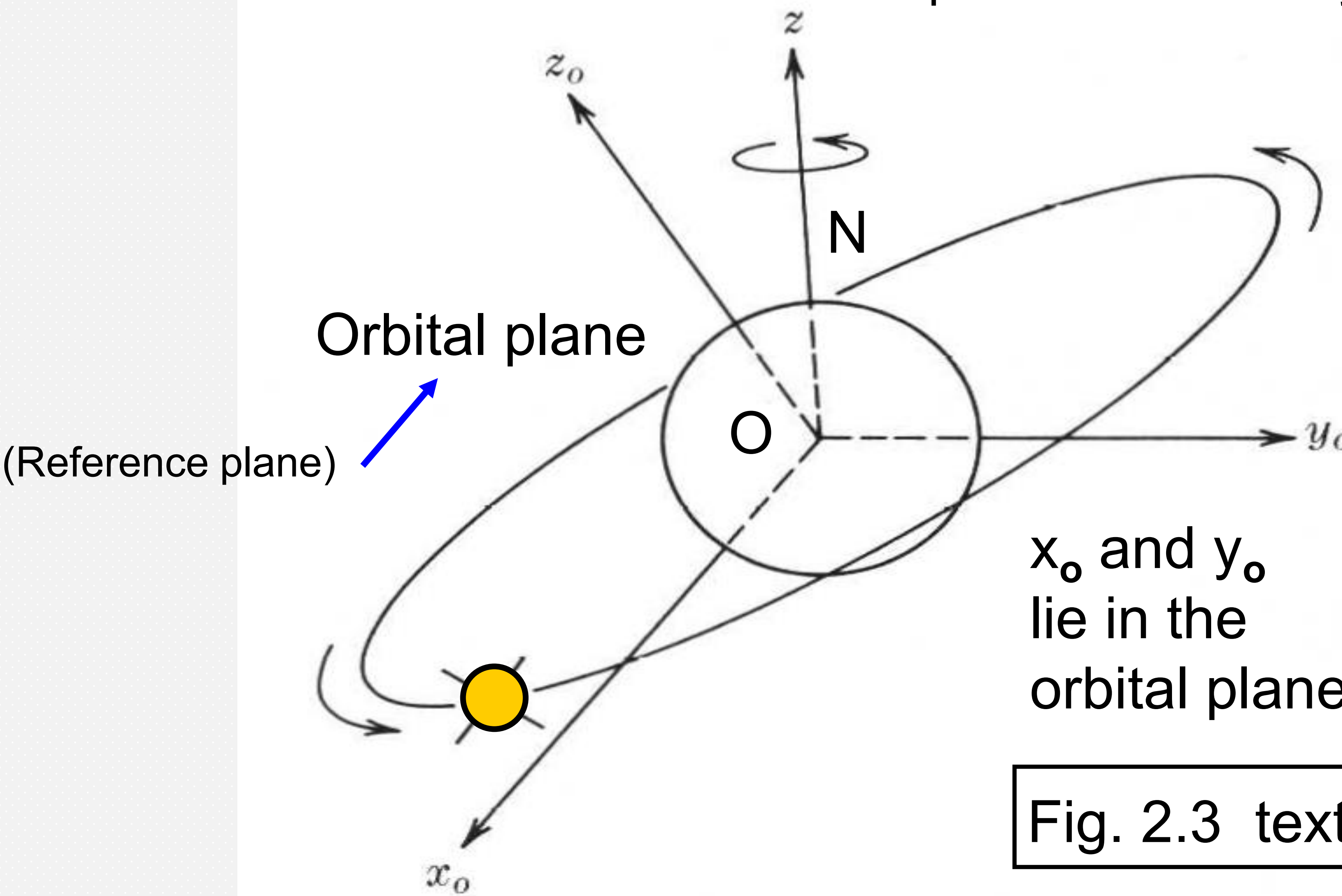
Satellite orbits

- Isaac Newton (1642 - 1727) (known for liking apples) derived Kepler's laws from his three laws of motion using integral and differential calculus
- It is believed that Newton invented integral and differential calculus while he was a student at Cambridge University in 1661-5
- Newton published his *Principia* in 1687 stating the laws of motion, law of universal gravitation, and a derivation of Kepler's laws

Satellite Orbits

- While Equation 2.10 is completely general, valid in any coordinate system…
- …it is easier to solve in a coordinate system that is based on the plane of the orbit
- *Orbital plane rectangular axes* are indicated by x_0 , y_0 , z_0 (see Sec 2.2)

Orbital plane coordinate system



x_o and y_o lie in the orbital plane

Fig. 2.3 text

Satellite Orbits

- Solution is easier still in *orbital plane polar coordinates* indicated by r_0 and ϕ_0
- These are easily related to the x_0 , y_0 , z_0 coordinates
- Solution of Equation 2.10 results in an expression for r_0 in terms of ϕ_0 - this is the *equation of the orbit*
- Equation of the orbit is an *ellipse*

Drawing an Ellipse

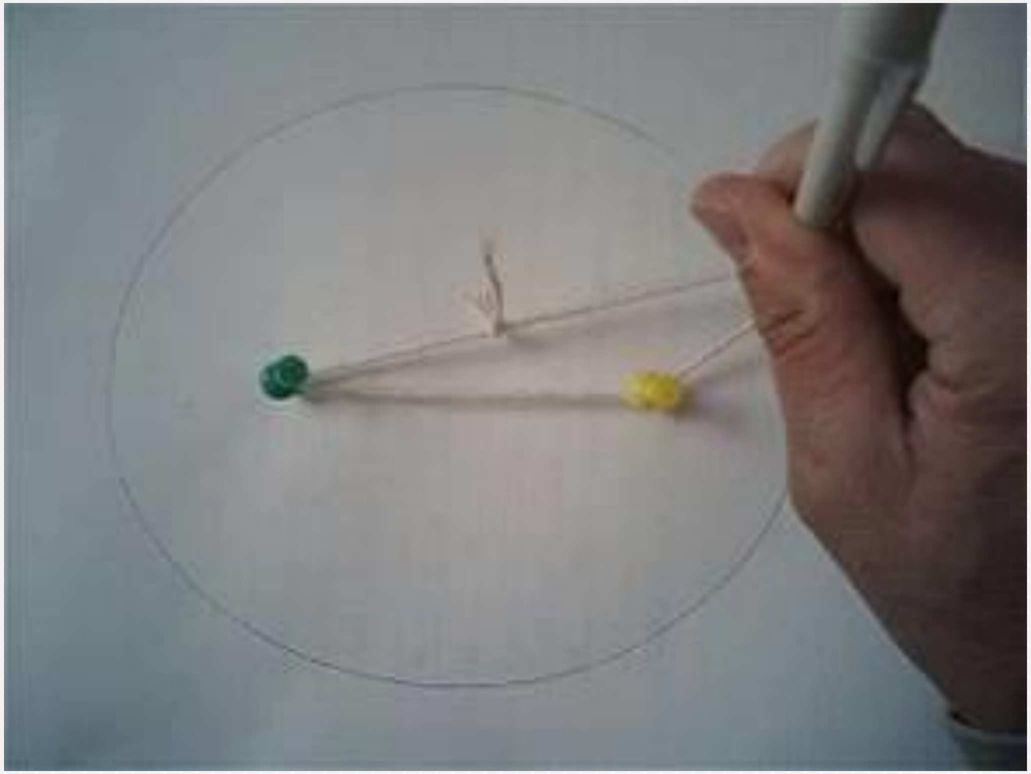
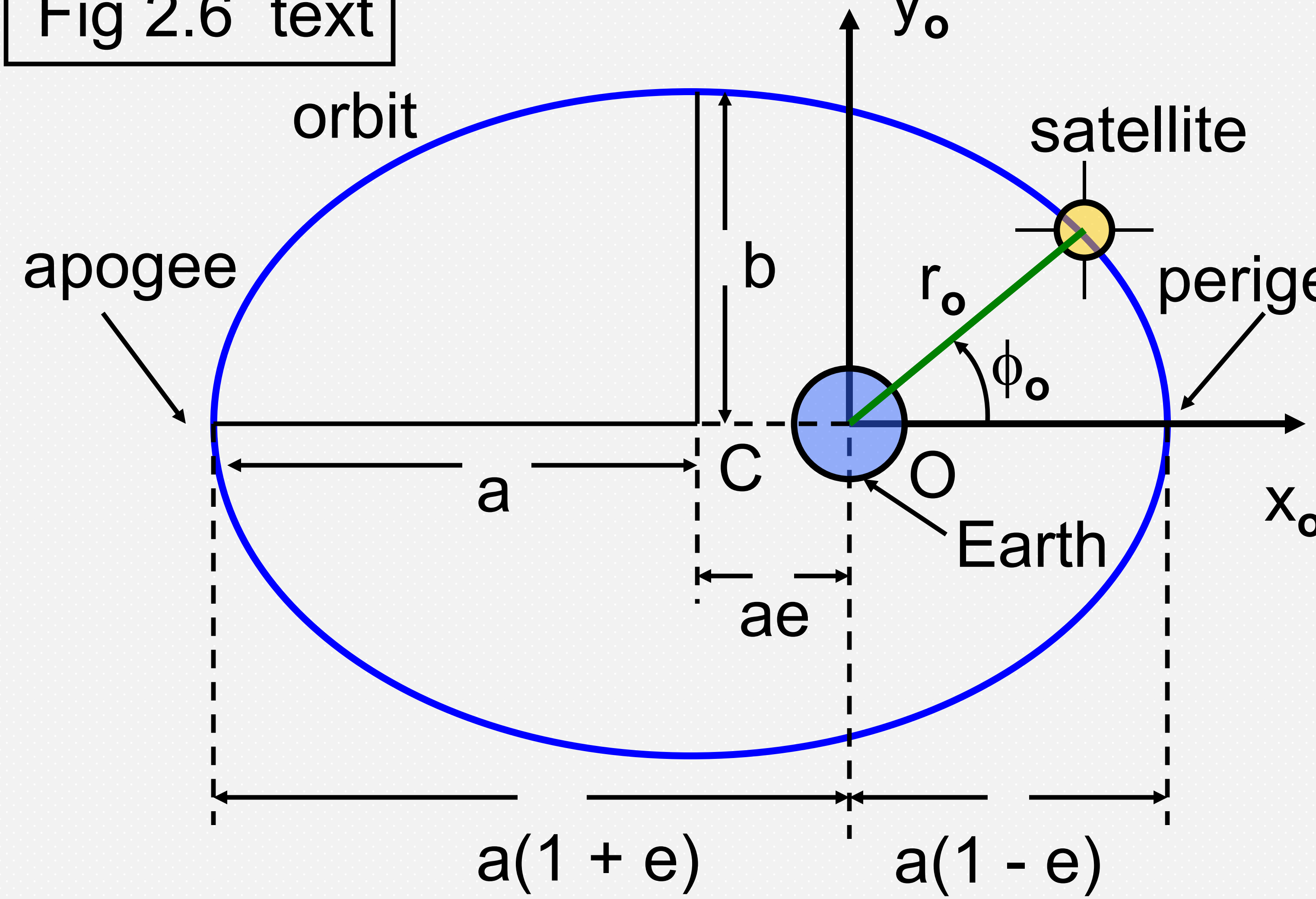


Fig 2.6 text



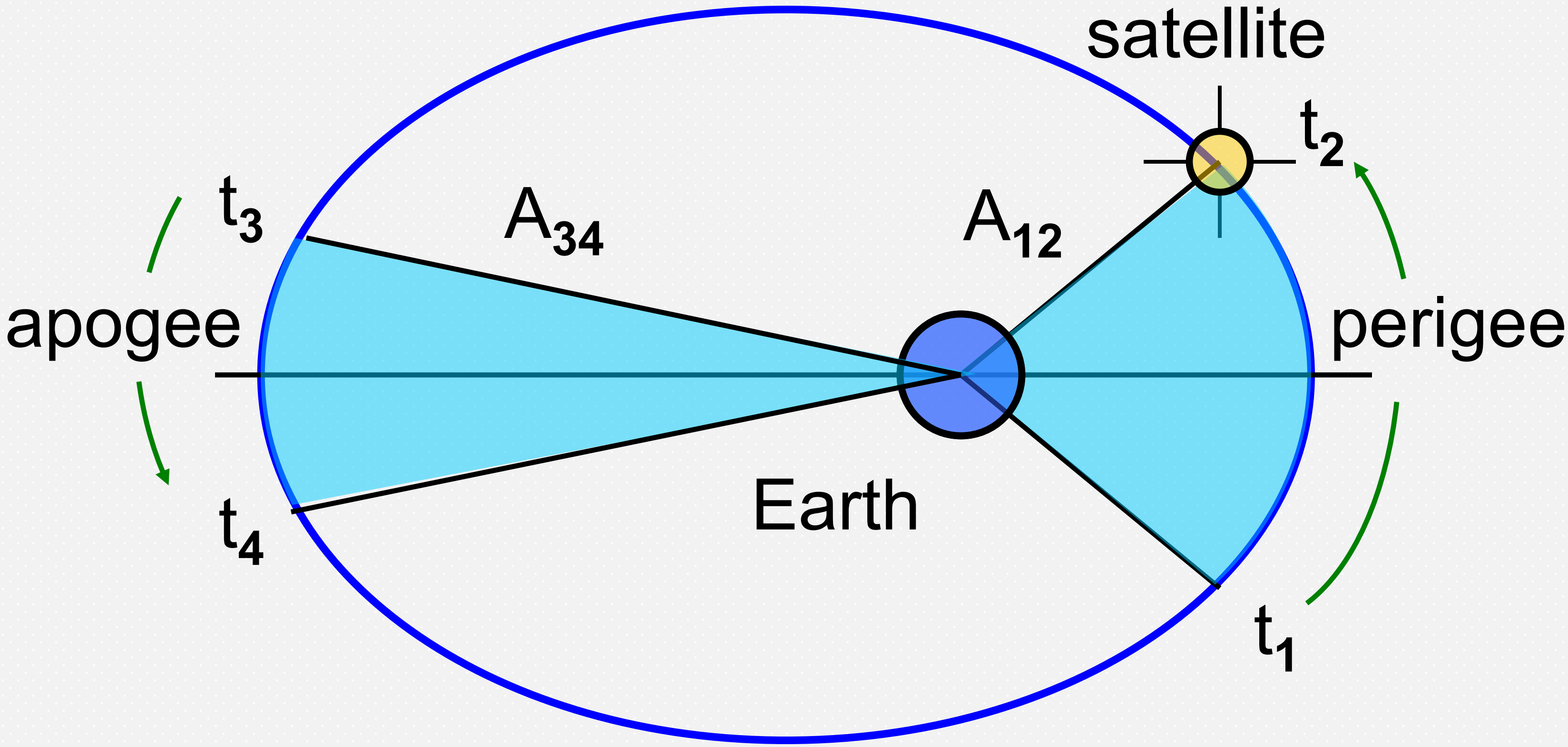
Kepler’s first law: Orbit is an ellipse

Satellite Orbits

- <u>Kepler's first law:</u>

  Orbit is an ellipse with Earth at one focus
- Other focus is empty
- The *eccentricity,* $e = (a^2 - b^2)^{1/2} / a$,defines shape of the ellipse and $0 \leq e \leq 1$
- Semi-major axis has length a
- Semi-minor axis has length b
- *Apogee* is point where satellite is most distant
- *Perigee* is point of closest approach

Fig. 2.5 text



Kepler's second law:

If : $t_2 - t_1 = t_4 - t_3$

then : $A_{12} = A_{34}$

Satellite Orbits

- <u>Kepler's second law:</u>
- Satellite sweeps out equal arc areas in equal times
- Satellite velocity is slowest at apogee
- Satellite velocity is highest at perigee
- Highly elliptical orbit means the satellite spends a long time near apogee

Orbital Period

Kepler's third law:

Orbital period is given by

$T^2 = (4\,\pi^2\,a^3) / \mu$ (Equation 2.21)

$\mu = 3.9860044 \times 10^5\ km^3/s^2$

(Kepler's constant)

The square of the period of revolution is proportional to the cube of the semi-major axis

Circular Orbit Example (2.1)

Find period of satellite in circular orbit at 322 km altitude:

Radius of orbit = $R_E + h$ = 6378.137 + 322
= 6700.137 km

$\mu = 3.986004418 \times 10^5 \ \ km^3/s^2$

$T^2 = (4\,\pi^2\,a^3) / \mu$ (a = radius of circular orbit)

$= 4 \times \pi^2 \times (6700.137)^3 / 3.986004418 \times 10^5 \ \ s^2$

$= 29790171.94 \ \ (\text{seconds})^2$

$T = 5{,}458.037$ s = 90 mn 58.037 sc

Elliptical Orbit Example (2.2)

- Find the period of orbit and eccentricity of a satellite with Perigee at 500 km height and Apogee at 39,152 km height
- For the period, we need to determine a, the length of the semi-major axis, to apply Kepler's third law
- The distance between apogee and perigee is

$$a(1 + e) + a(1 - e) = 2a$$

- See next figure

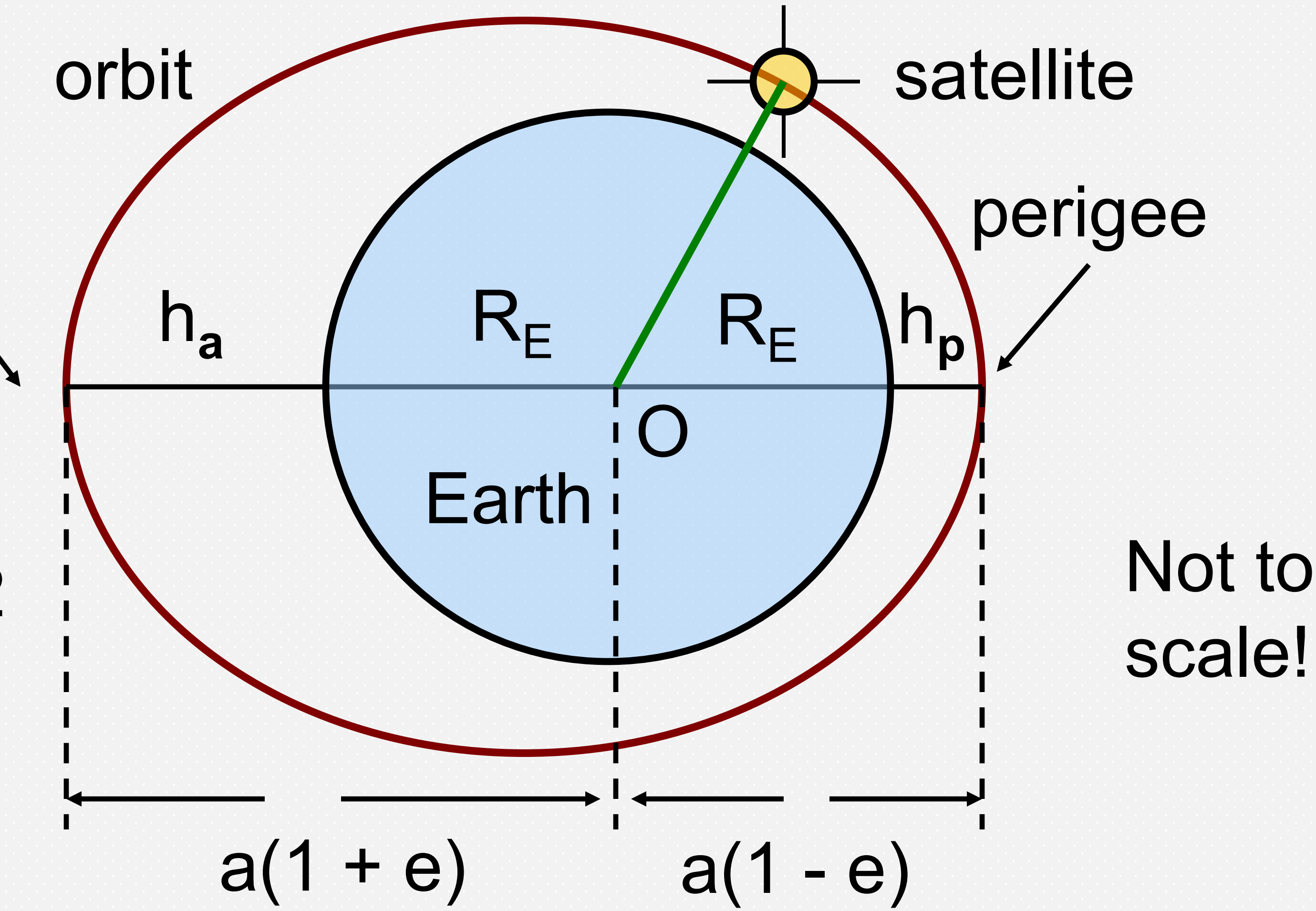


Fig Ex 2.2 (modified)

Example 2.2 h_a = 39,152 km, h_p = 500 km

Elliptical Orbit Example

- Taking Earth's mean radius to be 6378.137 km:

  $(39{,}152 + 6378.137) + (6378.137 + 500) = 2a$

- Hence $a = 26{,}304.137$ km

- Apply $T^2 = (4 \pi^2 a^3) / \mu$ (Eqn 2.21)

  $T^2 = 1{,}782{,}097{,}845.0 \ s^2$

  $T = 42{,}214.90075$ s

  $= 11$ hr 43 mn 34.9 sc

Elliptical Orbit Example

- The orbital eccentricity can be found by examining the distance to perigee (or apogee)

$$\text{Perigee distance} \quad = \quad a\,(1 - e)$$

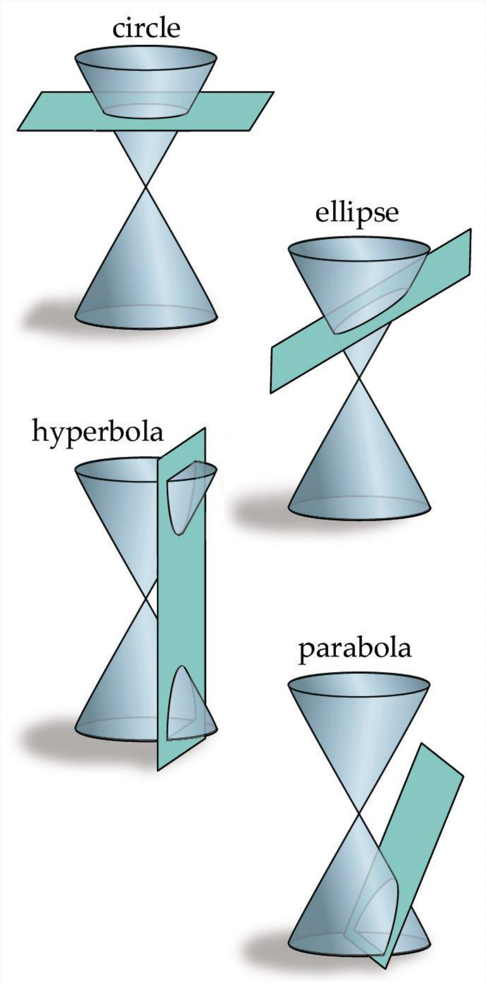
$$(6378.137 + 500) \quad = \quad 26{,}304.137\;\;(1 - e)$$

- Hence $\quad e \;=\; 1 - 6878.137 \,/\, 26{,}304.137$

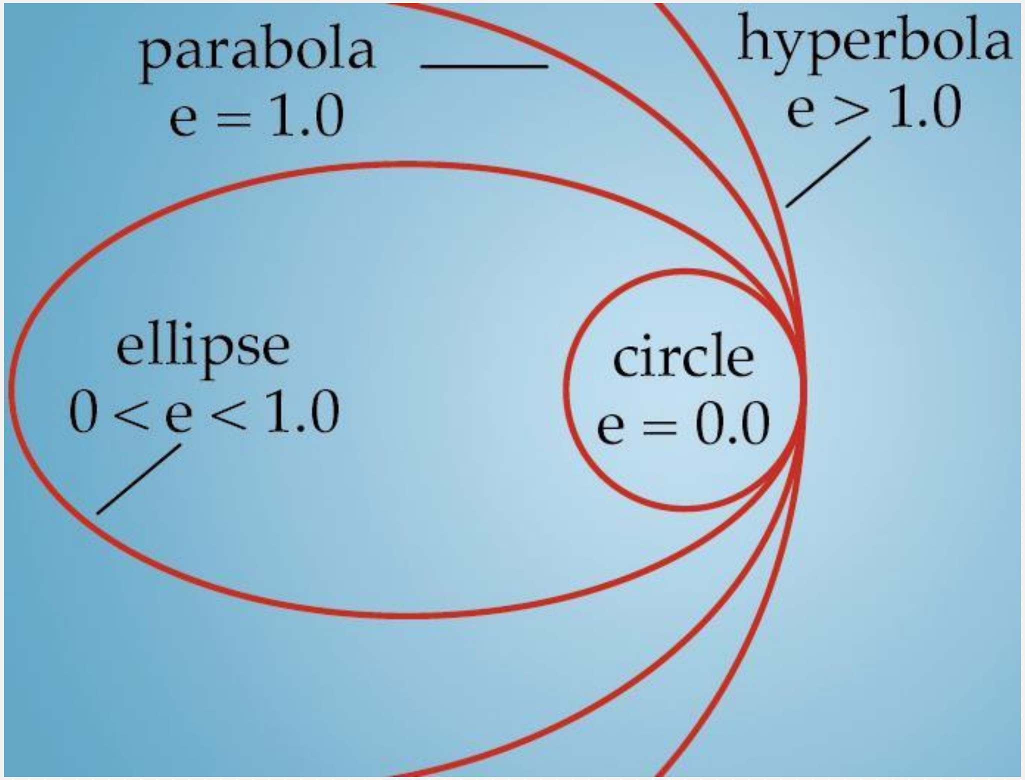
$$= \; 0.738515 \qquad [\text{dimensionless}]$$

Analyzing Orbits: Orbital Geometry

- A complete analysis of the satellite problem shows orbits can be any of the following shapes (sections of a cone):
  - Circle
  - Ellipse
  - Parabola
  - Hyperbola



Eccentricity, e, and possible ‘orbits’



- An encounter with e ≥ 1.0 does not result in a stable orbit, the satellite is lost

Useful Orbits: Geostationary Orbit

- In earth's equatorial plane, circular (i, e = 0)
- Radius of orbit, r = 42,164.57 km
- Orbital Period = 23 h 56 min. 4.091 s
= One *Sidereal Day*
- (Sidereal day = Earth's rotational period)
- Satellite appears to be stationary over a point on the equator to an observer

Note: Orbit radius = Earth radius + orbital height

Average radius of earth at equator = 6,378.137 km

Useful Orbits: Non-Geostationary

- Low Earth Orbit - LEO (250 - 1200 km)

  $T \approx$ 90 - 110 minutes

- Polar (Low Earth) Orbit (i = 90º)

  Earth rotates 22.5^{o} each orbit - useful for surveillance

- Sun Synchronous Orbit (SSO)

  Polar orbit aligned with the sun

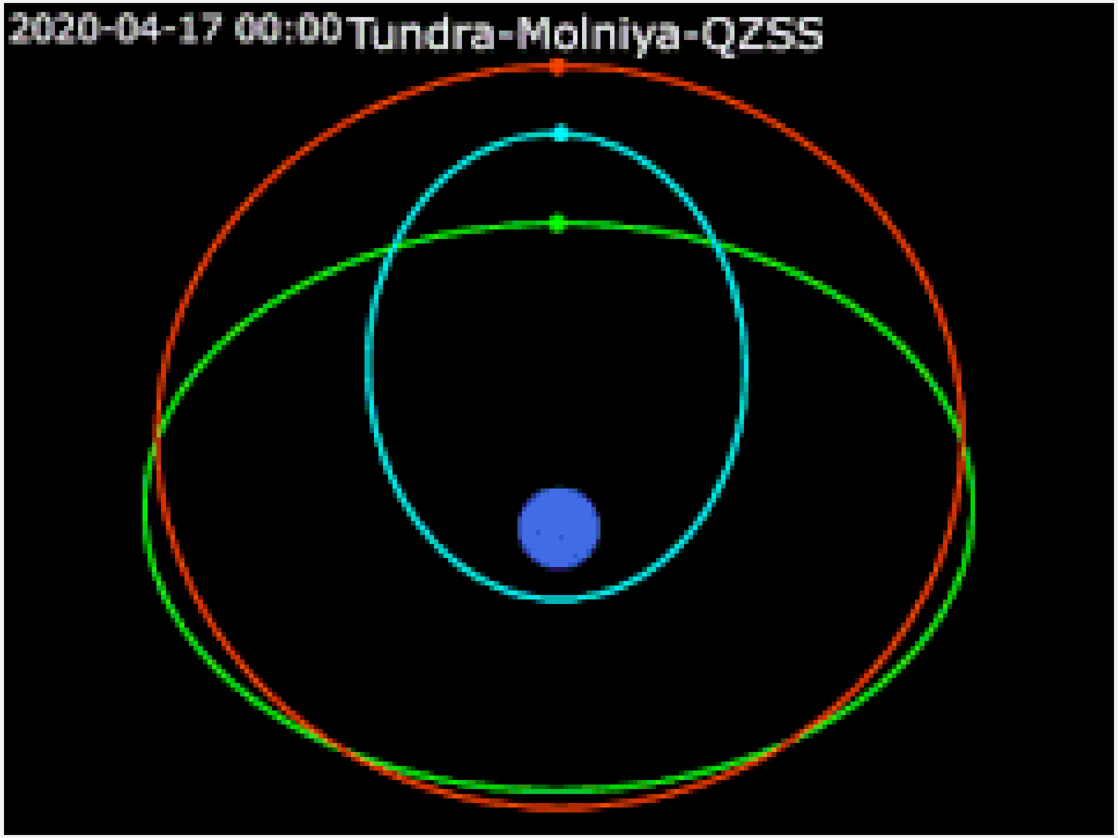
- Retrograde orbit (i > 90º)

  Orbits in direction opposite Earth's rotation

Useful Orbits: Molniya

- Molniya (Russian for lightning) orbit
- Highly elliptical
- $T \approx$ 11 hr 58 min
- Apogee 39,152 km
- Perigee 500 km
- Orbit track repeats every other orbit
- Satellite appears in same place in sky every 12 hours
- Used by former USSR to send TV to Siberia

Molniya orbit shown in blue (most elliptical)



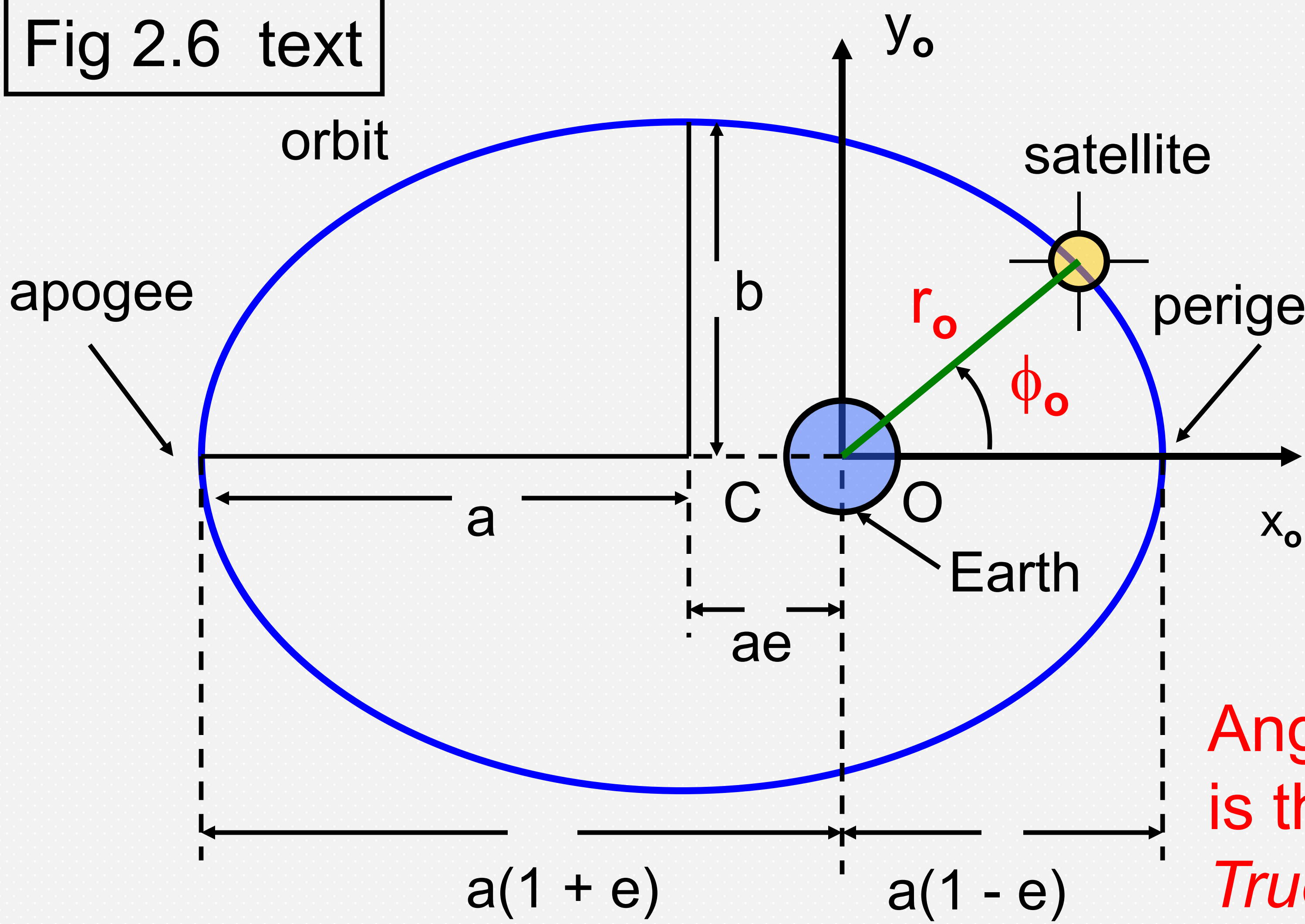
Location of satellite in orbital plane

- As shown in the text (Sec. 2.5) the equation of the orbit may be written in orbital plane polar coordinates as

$$r_o = \frac{a(1-e^2)}{1+e\cos(\varphi_o)} \qquad \text{Eqn 2.22}$$

- Note the two constants: *a* characterizes the size of the ellipse and *e* its shape

Fig 2.6 text

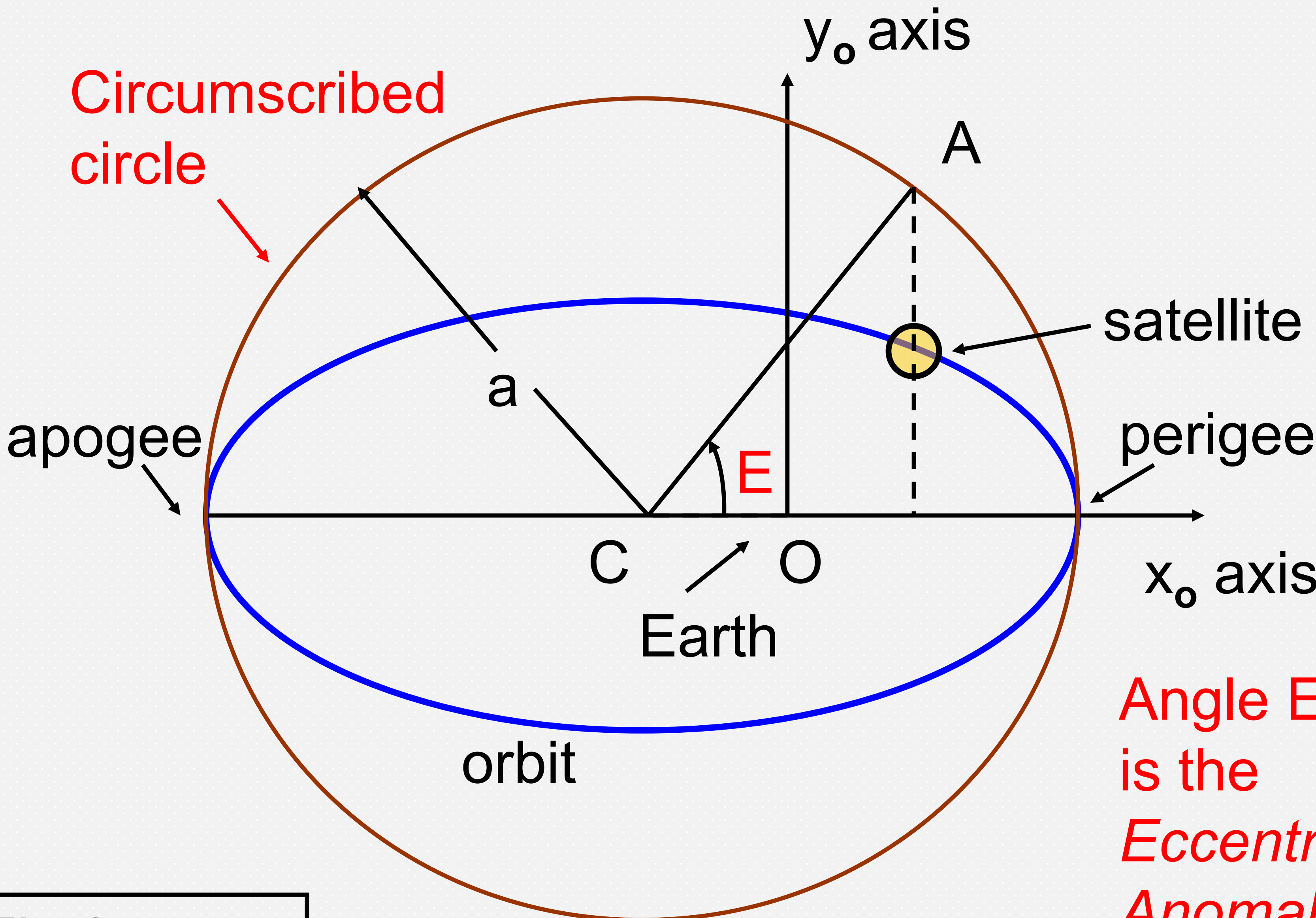


Angle ϕ_o is the *True Anomaly*

Location of satellite in orbital plane

- The polar angle φ_o is measured from the x_o axis and is called the *true anomaly*
- The term comes from astronomy
- The equation of the orbit gives the distance to the satellite, r_o , as a function of the true anomaly, φ_o , and

$$x_o = r_o \cos\varphi_o \qquad y_o = r_o \sin\varphi_o$$



Angle E is the *Eccentric Anomaly*

Fig 2.7 text

Location of satellite in orbital plane

- A *circumscribed* circle may be drawn to enclose the eclipse as shown in Fig 2.7
- A vertical line drawn through the satellite position intersects the circle at point A
- The angle between x_o and a line drawn from the center of the ellipse, C, to point A gives the *eccentric anomaly*, E

Location of satellite in orbital plane

- Information about the ellipse is provided by

  - Semimajor axis, *a* (size of the orbit)

  - Eccentricity, *e* (shape of the orbit)

  >> *first two orbital elements*

- The location of the satellite at any time *t* is referenced to the

  - Time of perigee, t_p

  >> *third orbital element*

Location of satellite in orbital plane

- The average angular velocity of the satellite in its orbit is given by (Eqn 2.25)

$$\eta = (2\pi)/T = (\mu^{1/2})/(a^{3/2}) \quad [\mathrm{rad/sec}]$$

- The *mean anomaly*, M, is given by

$$M = \eta(t - t_p) \quad [\mathrm{rad}]$$

- Mean anomaly is the arc length (in radians) the satellite would have traversed on the circumscribed circle from perigee moving at angular velocity η

Useful orbital parameters

- ***T*** *- period* (Eqn 2.21)
- η - average angular velocity
  and $\eta = 2\pi / T$ (Eqn 2.25)
- **M** - Mean Anomaly, distance traversed around circumscribed circle since perigee and $M = \eta\,(t - t_p)$ (Eqn 2.30)
- **E** - Eccentric Anomaly, related to **M** by

$$\boldsymbol{M = E - e\sin(E)} \quad \text{[rad]} \quad \text{(Eqn 2.30)}$$

Locating a Satellite in its Orbit

- The goal is to solve for (r_o , ϕ_o) or (x_o , y_o) for a time t knowing (a, e, t_p)
- Use the six-step procedure (p. 29)
- Given: an observation time t

  - calculate T, η, and M

  - iterate $M = E – e \sin(E)$ to find E [rad]

  - use ellipse relations to find (r_o, ϕ_o) from E

  - transform to (x_o , y_o) if desired

Locating a Satellite in its Orbit

- The ellipse relations for finding (r_o , ϕ_o) from *E* are

$$r_o = a(1 - e\cos E) \qquad \text{Eqn 2.26}$$

$$r_o = \frac{a(1-e^2)}{1+e\cos(\varphi_o)} \qquad \text{Eqn 2.22}$$

- We now know how to find the location of the satellite *in the orbital plane* at any time *t* using *three* orbital elements (*e*, *a*, t_p)

Locating the Satellite with respect to Earth

- To find the location of the satellite *in the sky*, we need to determine its location with respect to an observer on Earth (next section)
- Ultimately, to specify the absolute (inertial) coordinates of a satellite at time *t* we will need *six* orbital elements

** Breaking News **

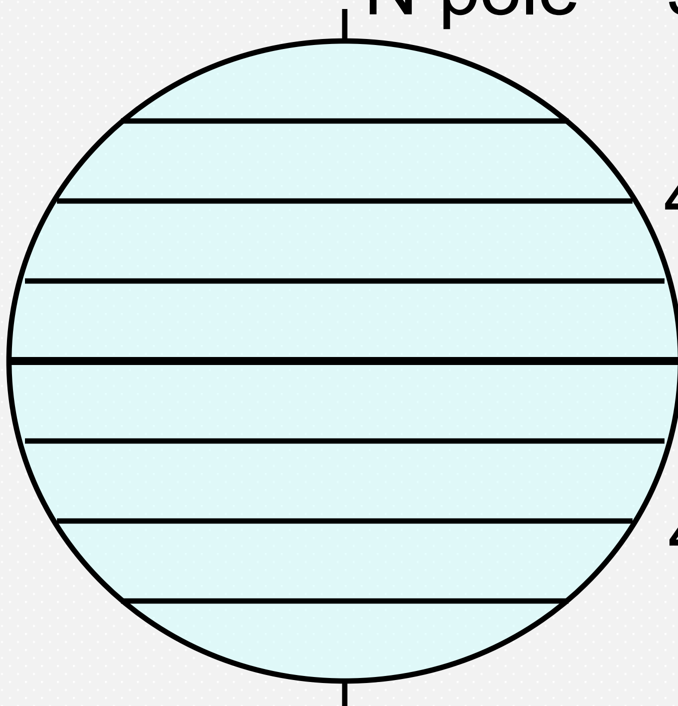
- The launch of Artemis II has been delayed until at least next month
- The testing on Monday evening (called a ‘wet dress rehearsal’) resulted in leaks of supercooled liquid hydrogen (LH2)
- Presumably the window for launching opens and closes with the position of the moon in its orbit

Artemis II at Launch Complex 39B (January, 2026)

Earth Station Coordinates

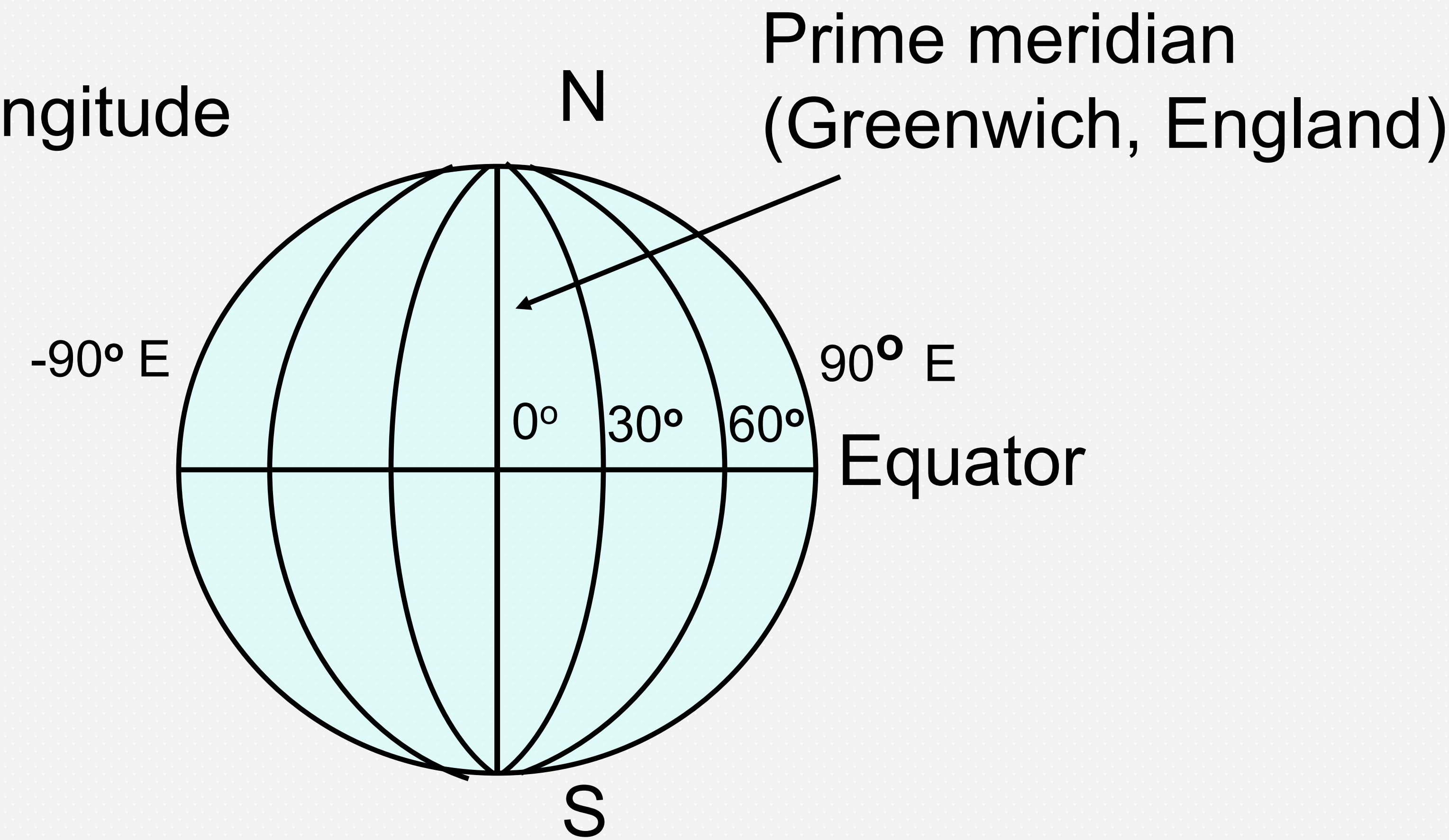
- Earth station coordinates are given by *Latitude* (L) and *Longitude* (l)
- Lines of longitude are *great circles* on the earth's surface, passing through both poles
- Lines of latitude are circles parallel to the equator
- Latitude and longitude are given as angles measured at the center of the earth

Latitude

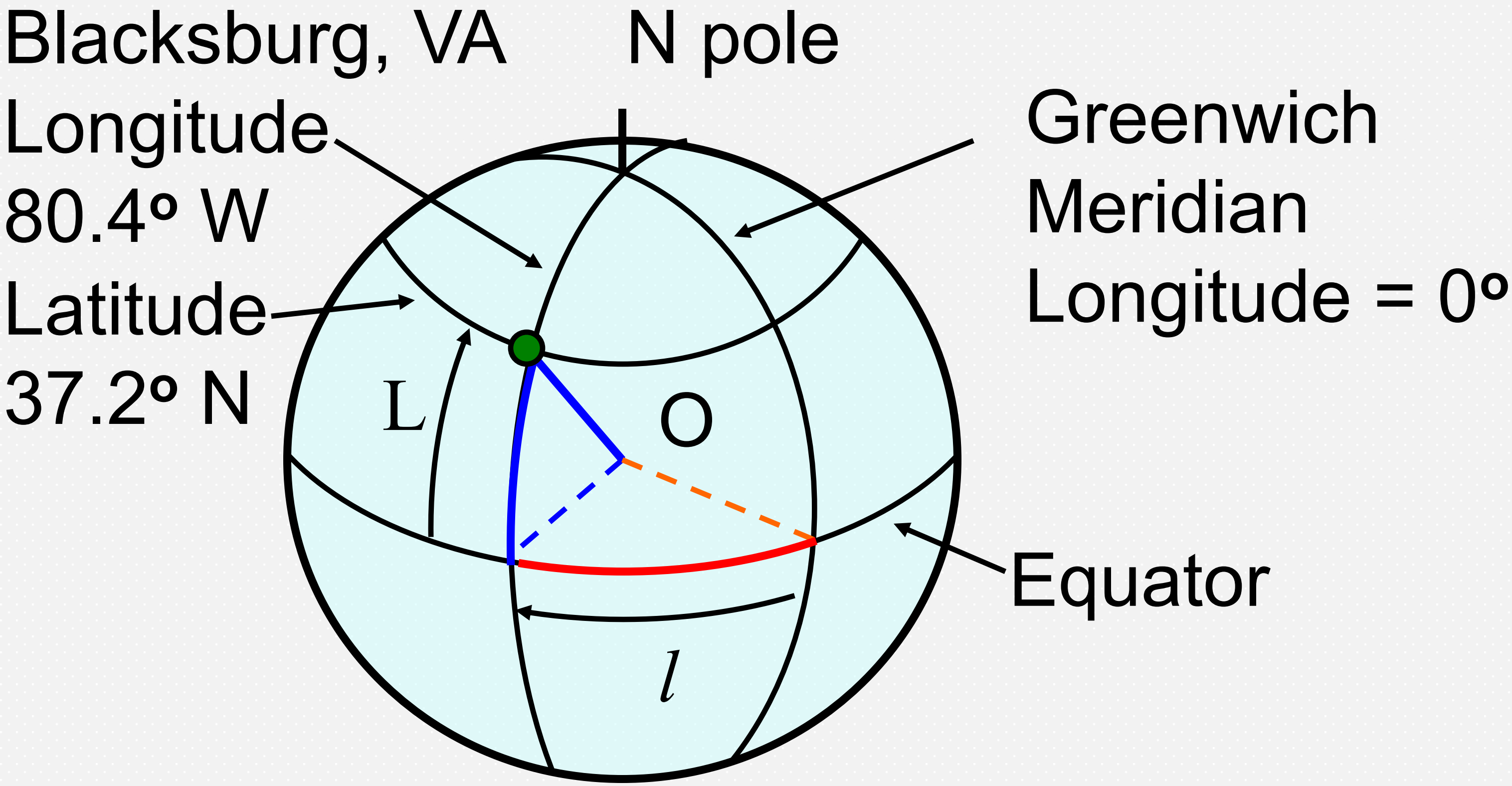


Lines of latitude are circles parallel to the equator.
Angles are measured from the center of earth.

Longitude



Lines of longitude are circles through both poles. Angles are measured at center of earth in the equatorial plane, relative to Greenwich meridian.



Latitude is defined N or S of equator.
Longitude is defined E or W of Greenwich meridian.
Blacksburg is at 37.2° N, 80.4° W

Where do we look in the sky?

- The goal is to find the *look angles* to the satellite from a position on Earth's surface
- We need to know the location of the satellite in the sky at a time t
- We already know how to locate the satellite in its orbit, i.e., determine (r_o, ϕ_o) or (x_o, y_o) using the first three orbital elements, a, e, and t_p
- Now we need to specify the location of the satellite with respect to an Earth station at (L, l)

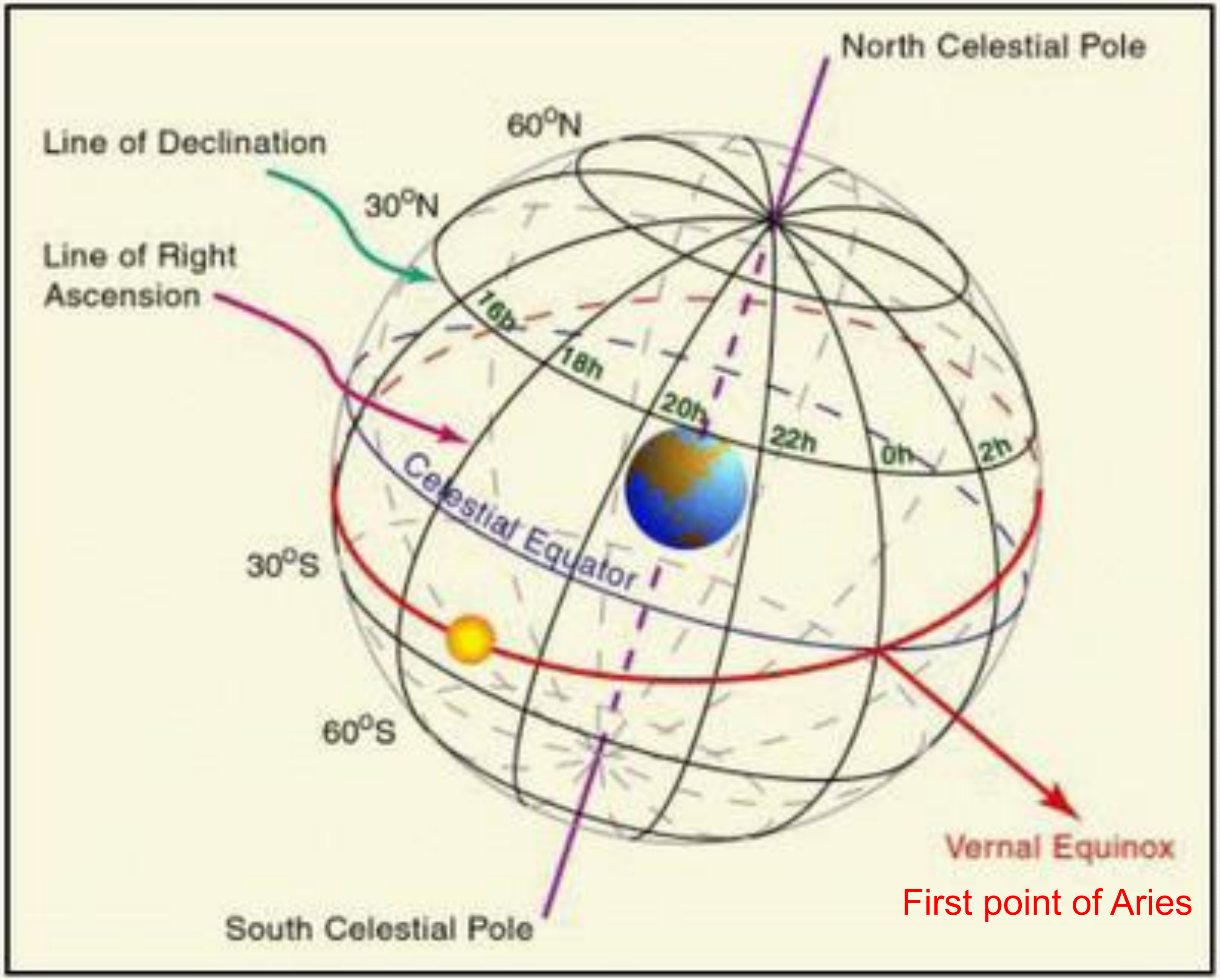
Look Angles

- To know where to look for the satellite in the sky, we must relate the orbital plane and time of perigee to the Earth
- First, let us consider how the positions of objects in the sky (stars, Sun, satellites) are specified

Sky Coordinates

- In exact analogy to longitude and latitude on the Earth, the sky is mapped with *right ascension* (RA) and *declination* (δ)

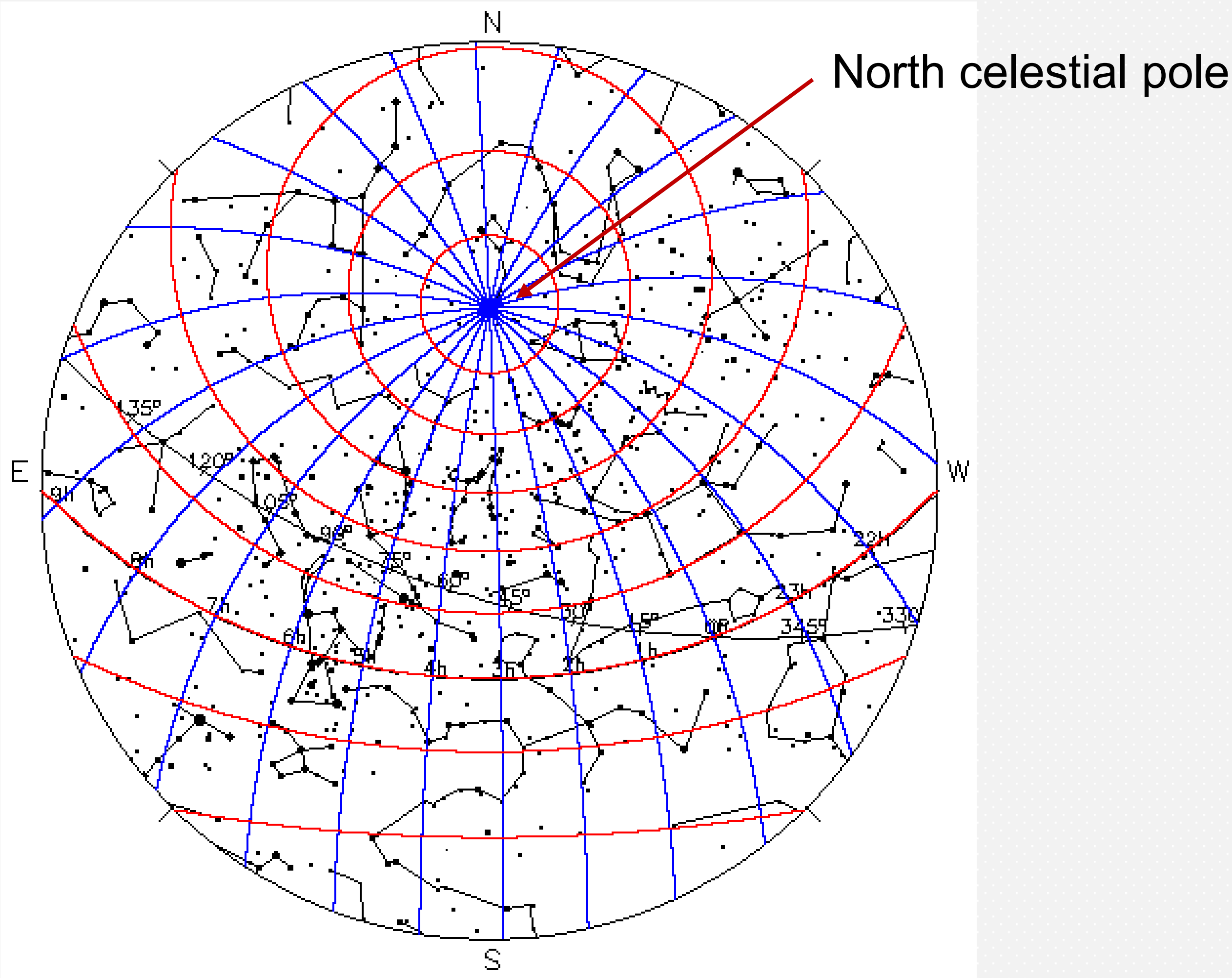
- Any sky object can be located from its (RA, δ) coordinates



First point of Aries

Sky Coordinates

- On the surface of the sky:
  - > right ascension corresponds to east longitude
  - > declination corresponds to north latitude
- Right ascension (RA) is expressed in degrees (0º - 360º) or hr:mn:sc (0 - 24 hr) with 1 hr = 15º
- Declination (δ) is expressed in degrees North or South with limits of 90º N and 90º S (= -90º N)



Star map showing lines of constant right ascension (blue) and constant declination (red) for a latitude of ~45º N at an instant of time

Mapping sky objects

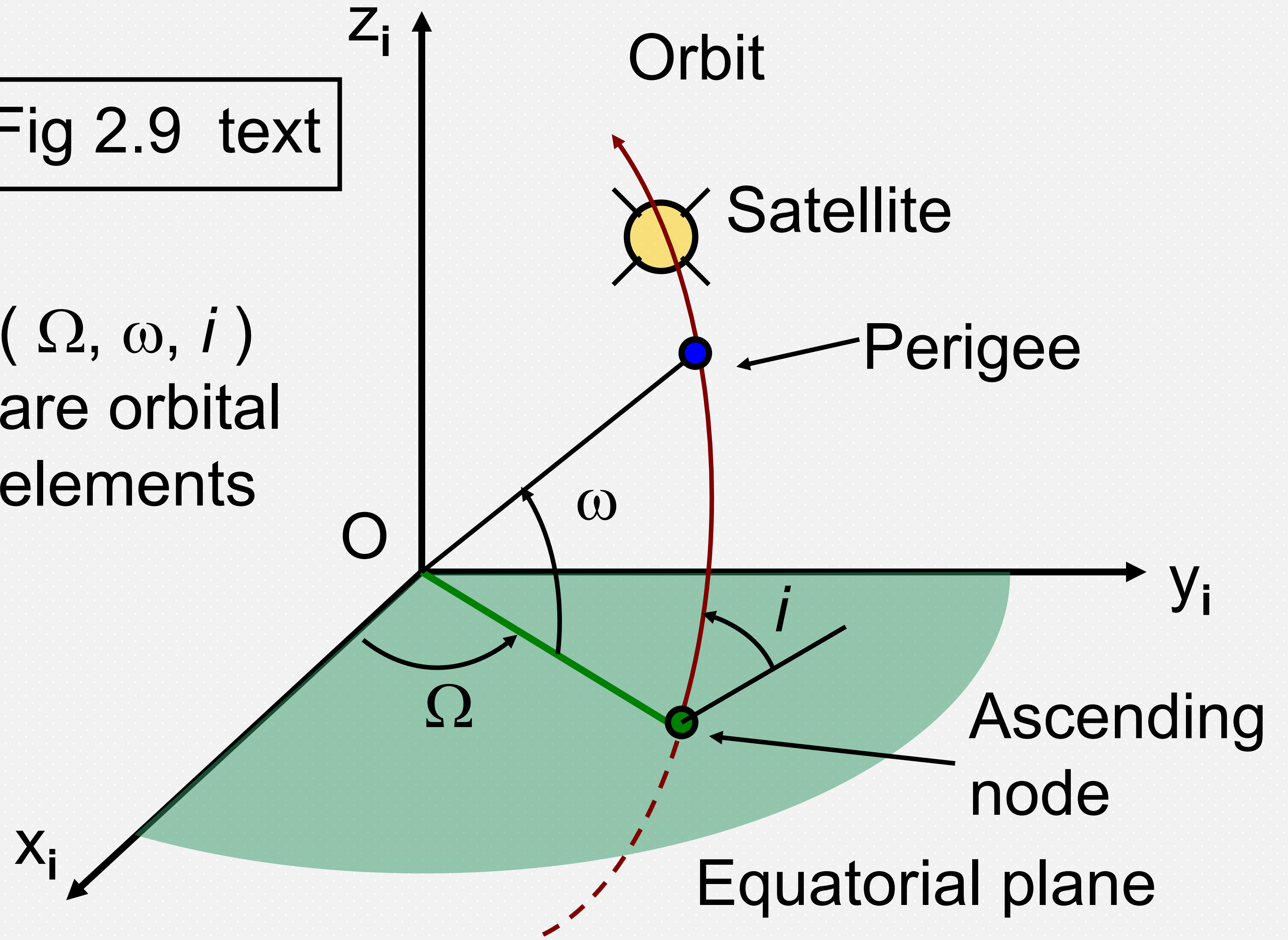
- The ‘equator’ in the sky where $\delta = 0$ is the projection of Earth’s equator and is called the *celestial equator*
- We will work in terms of the *geocentric equatorial coordinate system* defined by (x_i, y_i, z_i) with z_i parallel to Earth’s rotational axis
- The starting point for right ascension is the *first point of Aries* (RA = 0˚, $\delta = 0$). This is in a fixed direction relative to the distant stars and determines x_i

Describing an orbit with respect to Earth

- The satellite orbit is an ellipse lying in the orbital plane that can be *tilted* with respect to the equatorial plane
- The point where the satellite crosses the celestial equator moving northward can be *shifted* away from the first point of Aries
- The perigee can be *shifted* away from this crossing point

Fig 2.9 text

(Ω, ω, i) are orbital elements



Three additional Orbital Elements

- The angles in Fig 2.9 define the orientation of the orbit in geocentric coordinates
- The *ascending (descending) node* is where the satellite orbit passes through the equatorial plane travelling in a northerly (southerly) direction
- Angle i is the *inclination of the orbital plane*
- Angle Ω is the *right ascension of the ascending node*
- Angle ω is the *argument of the perigee*

Six Orbital Elements: a complete description

- Specification of the satellite orbit with respect to Earth requires six parameters - the orbital elements symbolized by (a, e, t_p, Ω, ω, i)
- This critical information is posted for many satellites as two lines of data at

http://celestrak.com/NORAD/elements/

NOAA 15 [+] (Timing info)

1 25338U 98030A 11025.26698606 .00000118 00000-0 69009-4 0 6853

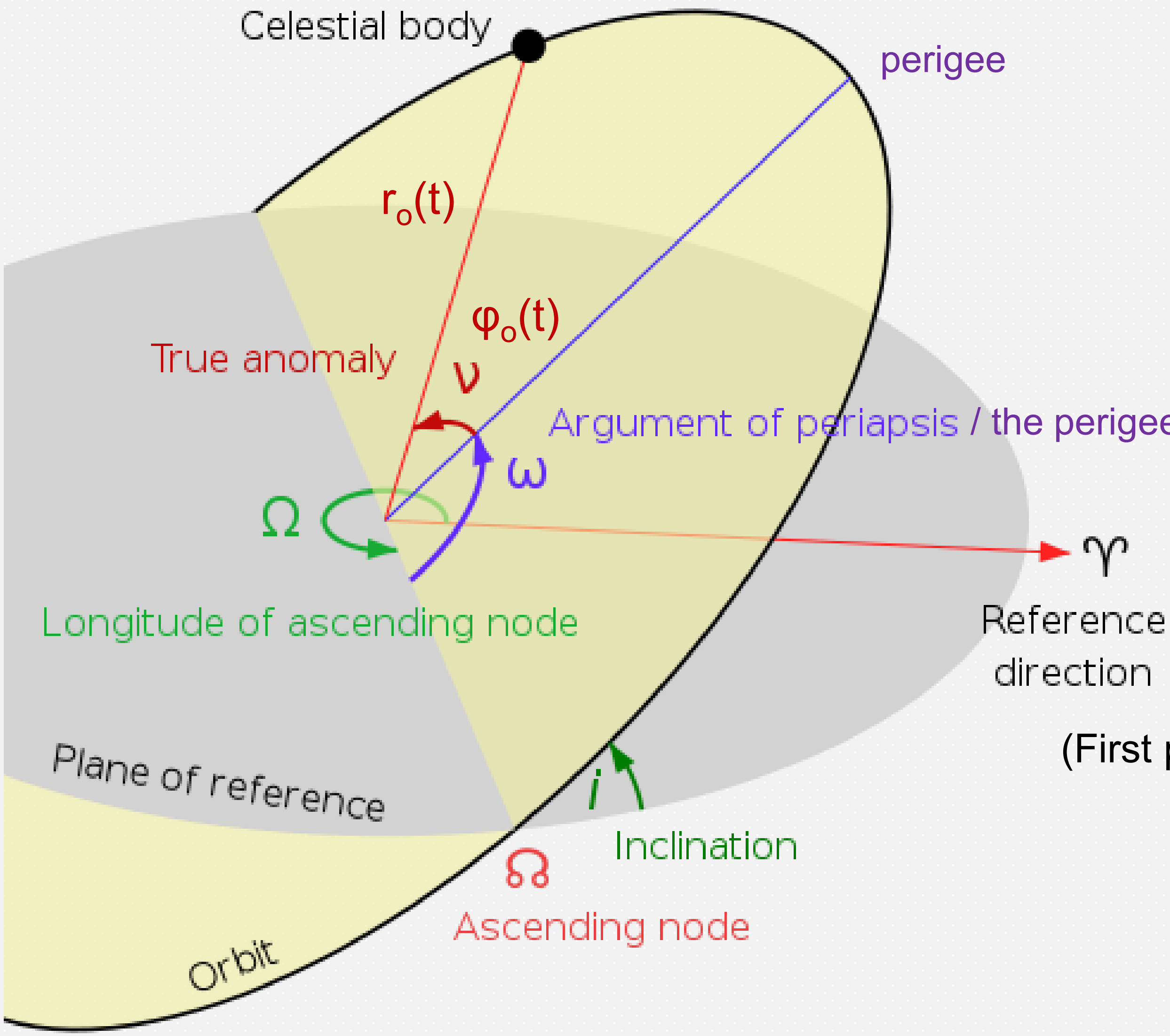
2 25338 98.6418 13.4695 0011171 144.6221 215.5707 14.24880073660374

(i Ω e ω M *Rev/day* |*Rev#*|)

Six Orbital Elements: Variations

- The NORAD 2-line posting does not state a value for *a*, the semi-major axis, but this can be inferred from the *Rev/day* parameter which gives the orbital period
- The 2-line posting also does not state a value for t_p , instead, it provides a value of mean anomaly (*M*) at a given time
- *M* can be combined with the period of the orbit to find t_p
- Six constants are needed to specify an orbit but the choice of constants is not fixed

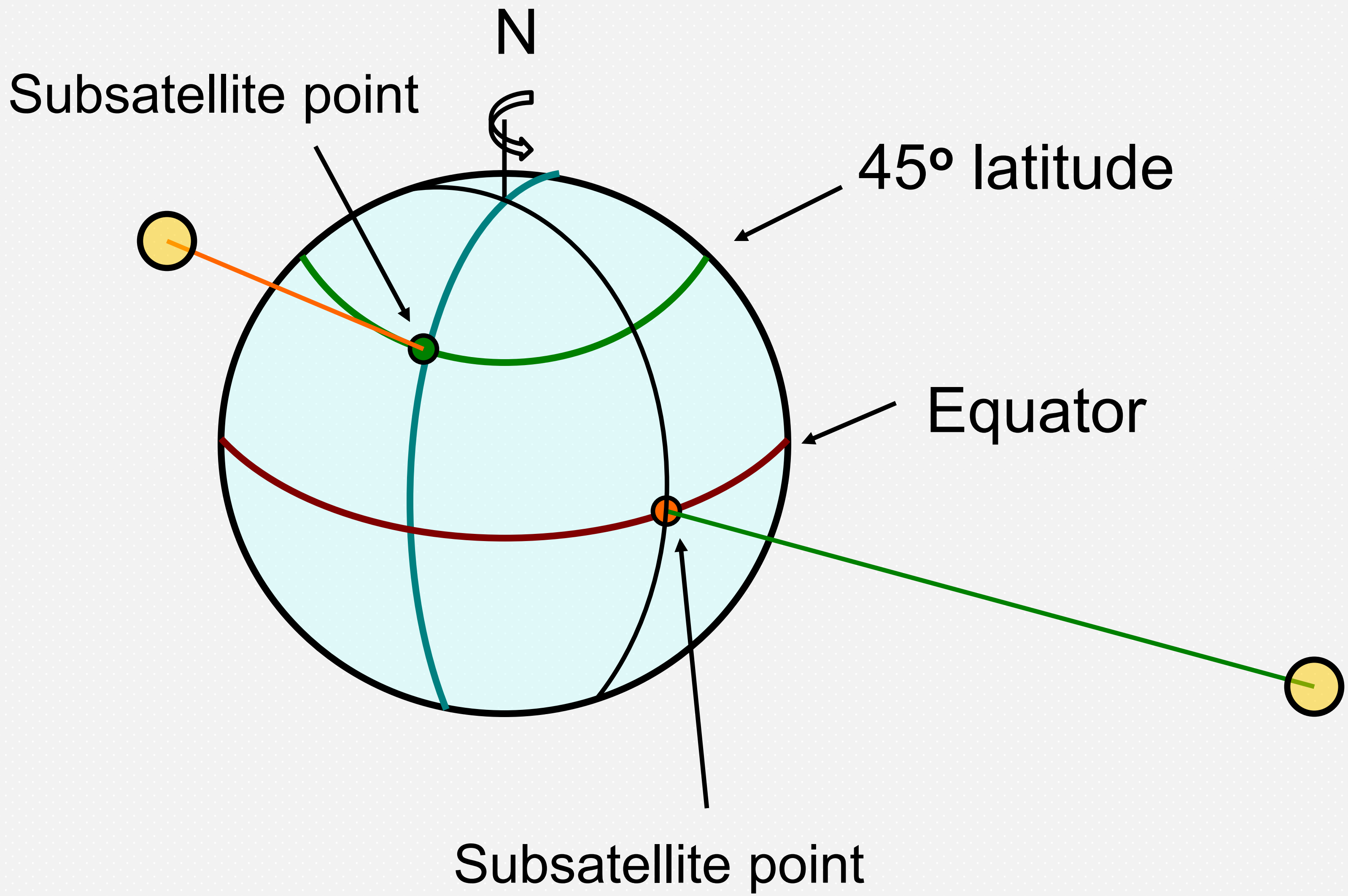
Orbital Elements



(First point of Aries)

Finding satellite position

- Given a set of orbital elements and a time *t* the position of a satellite can be determined
- Position is stated in terms of Right Ascension, RA, and declination, δ (like a star)
- A computer program is generally consulted to make the transformation
- The program will likely output the latitude and longitude of the point on Earth directly below the satellite


N
Subsatellite point
45° latitude
Equator
Subsatellite point

Subsatellite Point

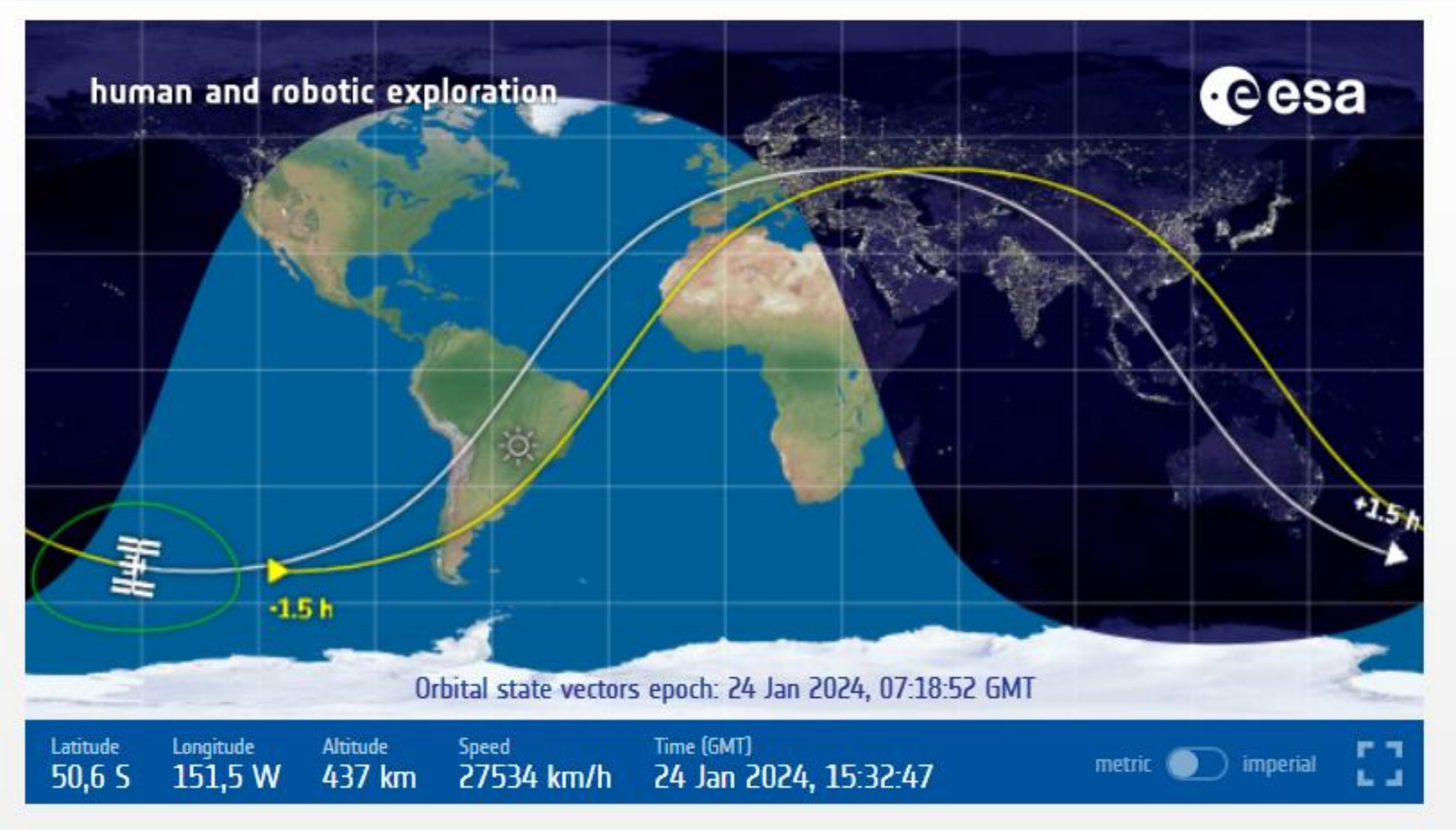
- A line connecting a satellite to the center of the earth passes through the earth's surface at the *subsatellite* point
- From the subsatellite point, the satellite is at the *zenith*
- From the satellite, the subsatellite point is at the *nadir*
- The value of *i* determines the highest latitude reached by the subsatellite point during an orbit

Satellite Ground Track

- Tracing the progress of the subsatellite point with time on a map produces the *ground track* of the satellite
- Example: NASA provides a real-time mapping of the ground track of the ISS (i = 51.6º)

Live Space Station Tracking Map

https://www.nasa.gov/spot-the-station/



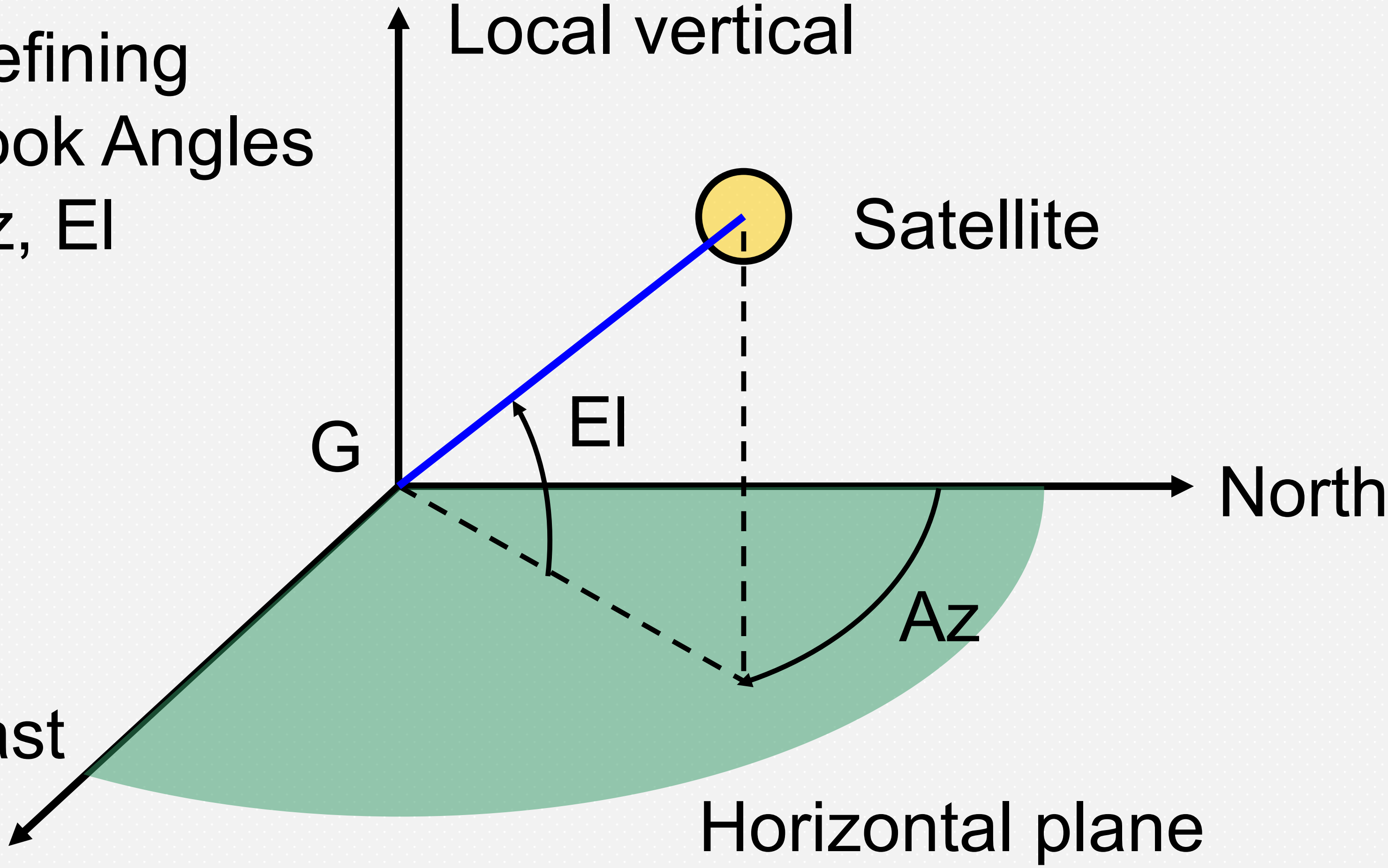
24 January, 2024 15:32:47 UT

Look Angles

- We generally want to know where to look in the sky to direct an antenna
- The desired ‘look angles’ are azimuth and elevation in local horizon coordinates

Fig 2.10 text

Defining
Look Angles
Az, El



Look Angles for Satellites

- Look angles: Azimuth (Az) and Elevation (El)
- Az is measured wrt the projection of the pointing direction (vector) onto the plane of the local horizon..

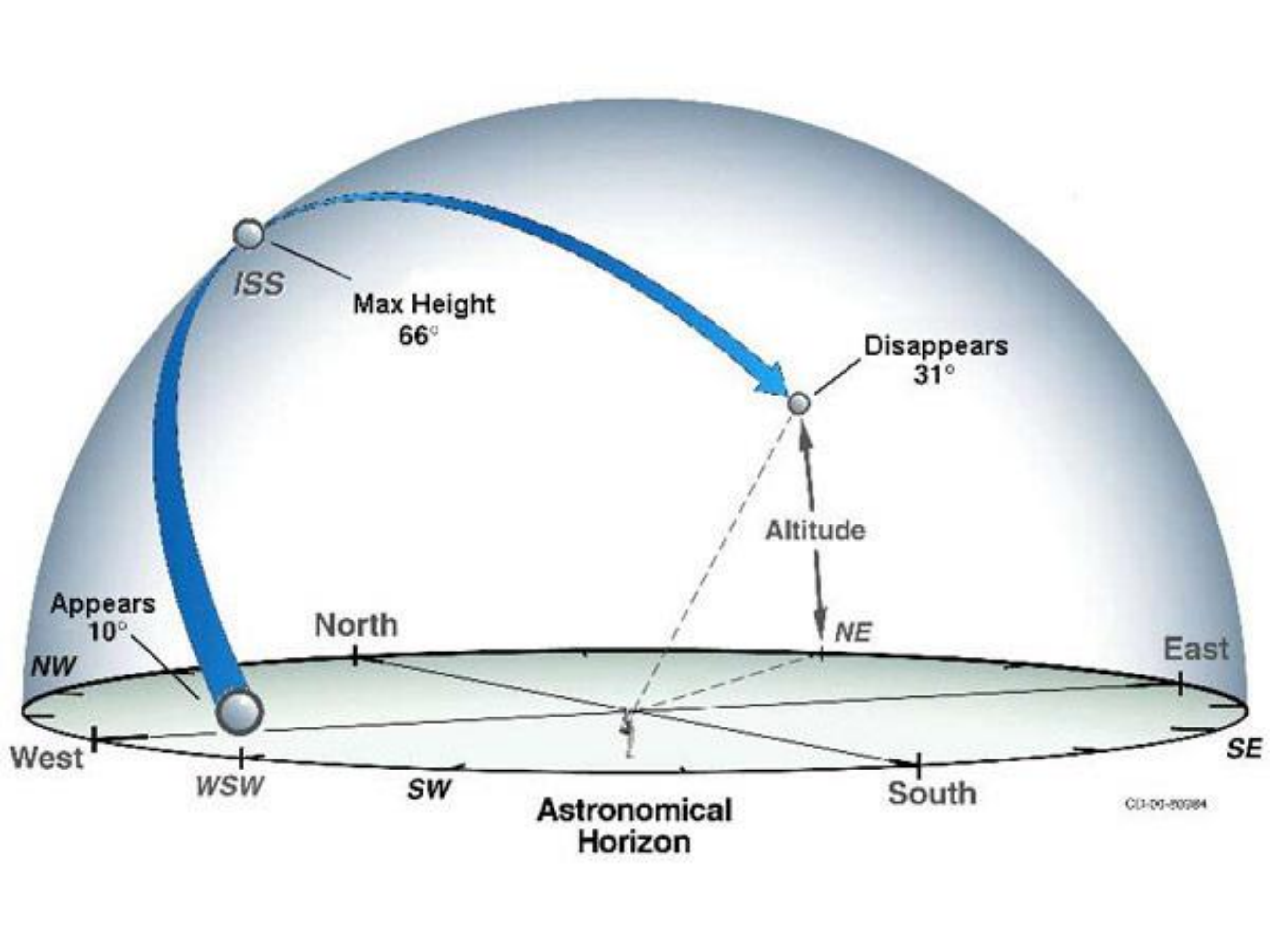
  .. and from N in the direction of E (0 - 360°)

- El is measured upwards from the plane of the local horizon (0 - 90°)

Look Angles for Satellites

- In addition to the tracker map, NASA used to provide a Table of look angles for visible passes of the ISS as viewed from a given Earth Station
- For example, from Blacksburg, Arlington
- The NASA calculator worked though the equations and converted satellite position in sky coordinates (RA, δ) to local horizon coordinates (Azimuth and Elevation)
- Then you would know where to look…

Tracking the ISS in the evening/morning sky



ISS Sighting Opportunities for Blacksburg VA, 2025

(Note: deprecated on the NASA website – replaced by an app)

	Date	Visible	Max Height*	Appears	Disappears	Share Event
	Sun Jan 26, 7:00 PM	2 min	16°	10° above NNW	16° above NNE	
	Mon Jan 27, 6:11 PM	3 min	12°	10° above N	10° above NE	
	Mon Jan 27, 7:47 PM	1 min	18°	10° above NW	18° above NW	
	Tue Jan 28, 6:58 PM	3 min	32°	10° above NNW	32° above NE	
	Wed Jan 29, 7:45 PM	2 min	33°	10° above WNW	33° above W	
>	Thu Jan 30, 6:56 PM	5 min	88°	10° above NW	22° above SE	
	Fri Jan 31, 7:45 PM	4 min	14°	10° above W	11° above SSW	
	Sat Feb 1, 6:56 PM	6 min	27°	10° above WNW	10° above S	

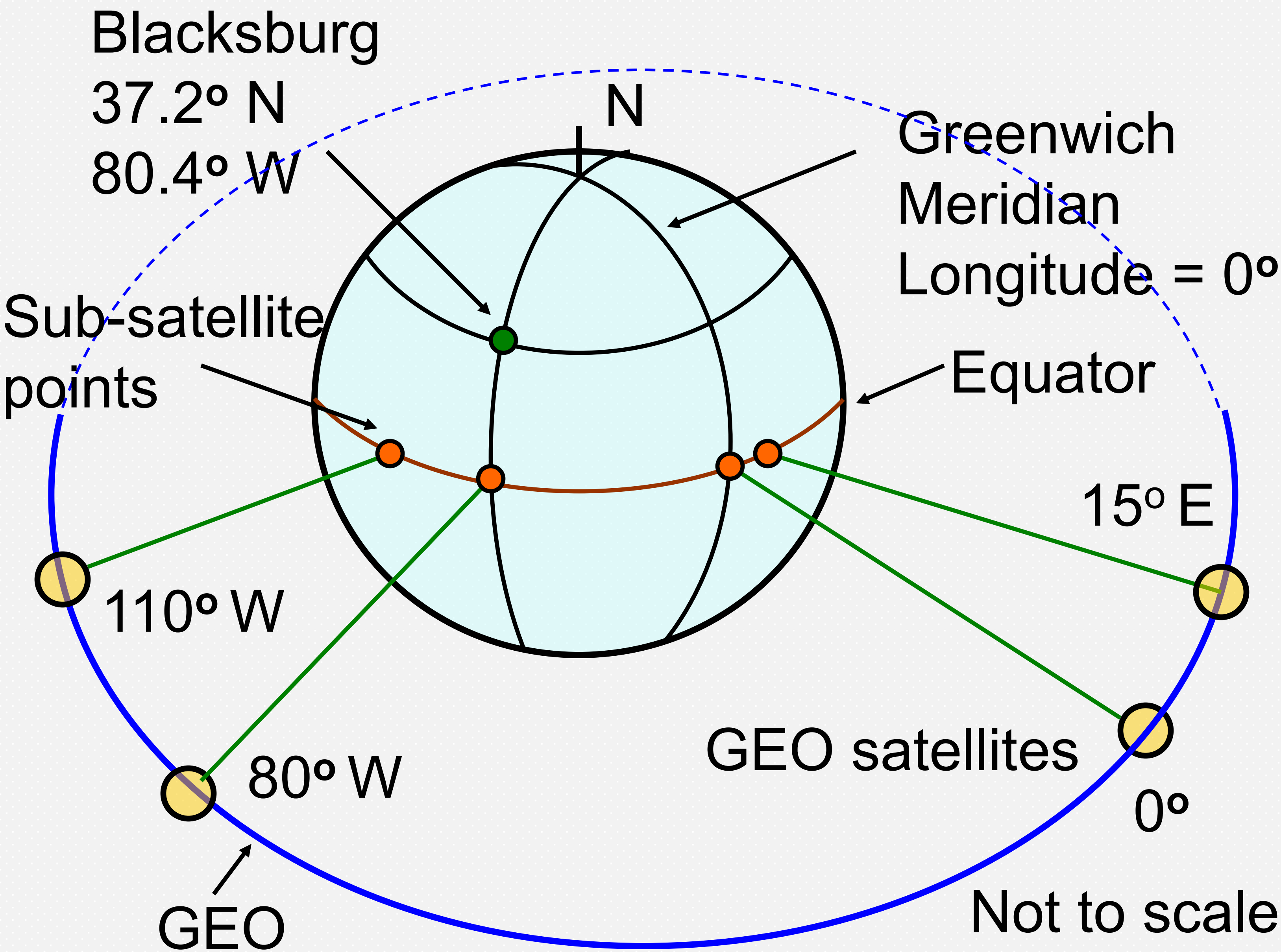
https://www.nasa.gov/spot-the-station/

Look angles for Satellites

- For satellites in LEO and MEO orbits, computer programs are usually consulted to solve for look angles
- The package with the longest heritage is the *Systems Tool Kit* (STK) from Analytical Graphics, Inc., (originally *Satellite Tool Kit*)
- We will work through the look angle calculations only for the case of GEO orbit

GEO Subsatellite Point

- Geostationary satellites are located in Earth's equatorial plane
- The subsatellite point of a GEO satellite is defined as a longitude – measured E or W of 0°
- Internationally, subsatellite longitude is measured from 0° through east
- In the North American sector, longitude is often measured from 0° through west
(80° W = 280° E)


Blacksburg
37.2° N
80.4° W
N
Greenwich
Meridian
Longitude = 0°
Sub-satellite
points
Equator
15° E
110° W
GEO satellites
80° W
0°
GEO
Not to scale

Look Angle Calculation

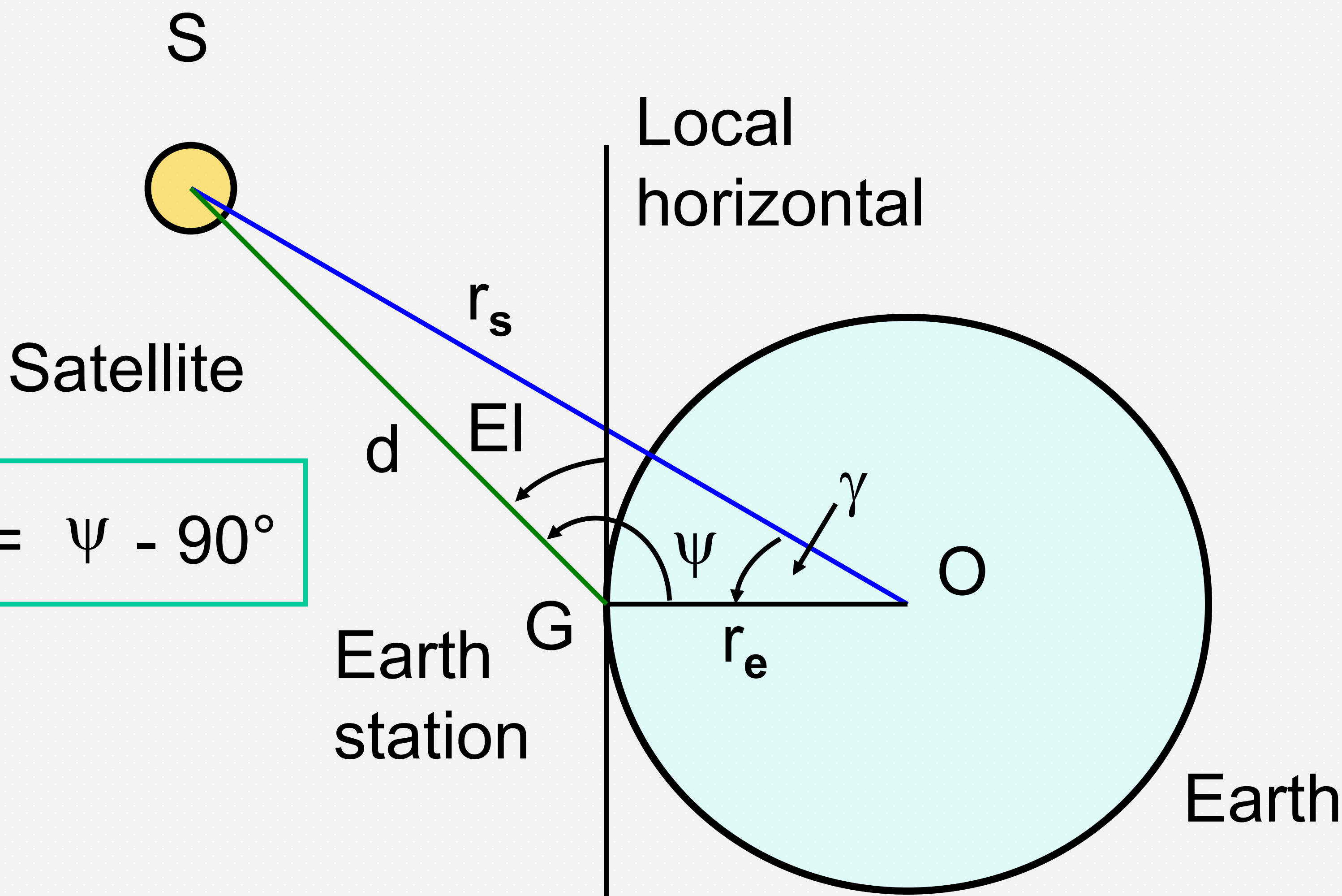
- Satellite location is given by the sub-satellite point, latitude L_s and longitude l_s
- A geostationary satellite has $L_s = 0^o$
- We must know the sub-satellite point accurately
- Earth station location is given by latitude L_e and longitude l_e

Calculating Look Angles

- Define angle at center of Earth as γ (gamma)
- Fig 2.12 shows geometry in plane defined by the satellite, earth station and Earth center
- Elevation angle is given by $El = \psi - 90^{\circ}$
- From spherical triangle geometry

$$\cos\gamma = \cos(L_e)\cos(L_s)\cos(l_s - l_e) + \sin(L_e)\sin(L_s)$$

Eqn 2.31



$$El = \psi - 90°$$

Fig 2.12 text

Geometry for El angle calculation

Calculating Look Angles

- For a truly geostationary satellite, L_s = 0º
- Then cos (L_s) = 1 and

$$\cos\gamma = \cos(L_e)\cos(l_s - l_e) \qquad \text{Eqn 2.36}$$

- Note that $(l_s - l_e)$ is longitude difference angle
- This angle is positive if longitude increases in an eastward sense from l_e to l_s
- Exercise caution when the satellite and earth station are on opposite sides of the prime meridian (0º longitude)

Elevation Angle Calculation

By the sine law we have

$$\frac{r_s}{\sin(\psi)} = \frac{d}{\sin(\gamma)} \qquad \text{Eqn 2.34}$$

And

$$\cos(El) = \frac{\sin(\gamma)}{\left[1 + \left(\frac{r_e}{r_s}\right)^2 - 2\left(\frac{r_e}{r_s}\right)\cos(\gamma)\right]^{1/2}} \qquad \text{Eqn 2.35}$$

Elevation Angle Calculation

- Then $\cos(El) = r_s \sin\gamma / d$
- d is the distance from the earth station to the satellite and is given by the denominator of Eqn 2.35
- Radius of the earth $r_e = 6378.137$ km
- Radius of GEO $r_s = 42{,}164.17$ km
- Then (Eqn 2.37; more precise form in text)

$$d = 42{,}164\,[1.022882 - 0.302538 \cos\gamma\,]^{1/2} \text{ km}$$

Elevation Angle Calculation

- This gives (Eqn 2.38)

  $\cos(El) =$

  $\sin\gamma / [1.022882 - 0.30253825 \cos\gamma]^{1/2}$

- An easier expression to evaluate is Eqn 2.39

  $El = \tan^{-1} [(6.6107345 - \cos\gamma) / \sin\gamma] - \gamma$

- For GEO satellites visible from the US, the distance d is often taken to be 38,500 km (approximation)

Azimuth Angle Calculation

- There are two steps to the Azimuth angle calculation
  1. Find the intermediate angle α
  2. Find Az from α and the relative positions of the satellite and earth station
- Equation 2.40 gives α

$$\alpha = \tan^{-1} [(\tan |(l_s - l_e)|) / \sin L_e]$$

- Now find Az from α and one of two cases

Azimuth Angle Calculation

Case 1: Earth station in the Northern Hemisphere

(a) Satellite to the SE of the earth station:

$Az = 180^{o} - \alpha$

(b) Satellite to the SW of the earth station:

$Az = 180^{o} + \alpha$

Case 2: Earth station in the Southern Hemisphere

(c) Satellite to the NE of the earth station:

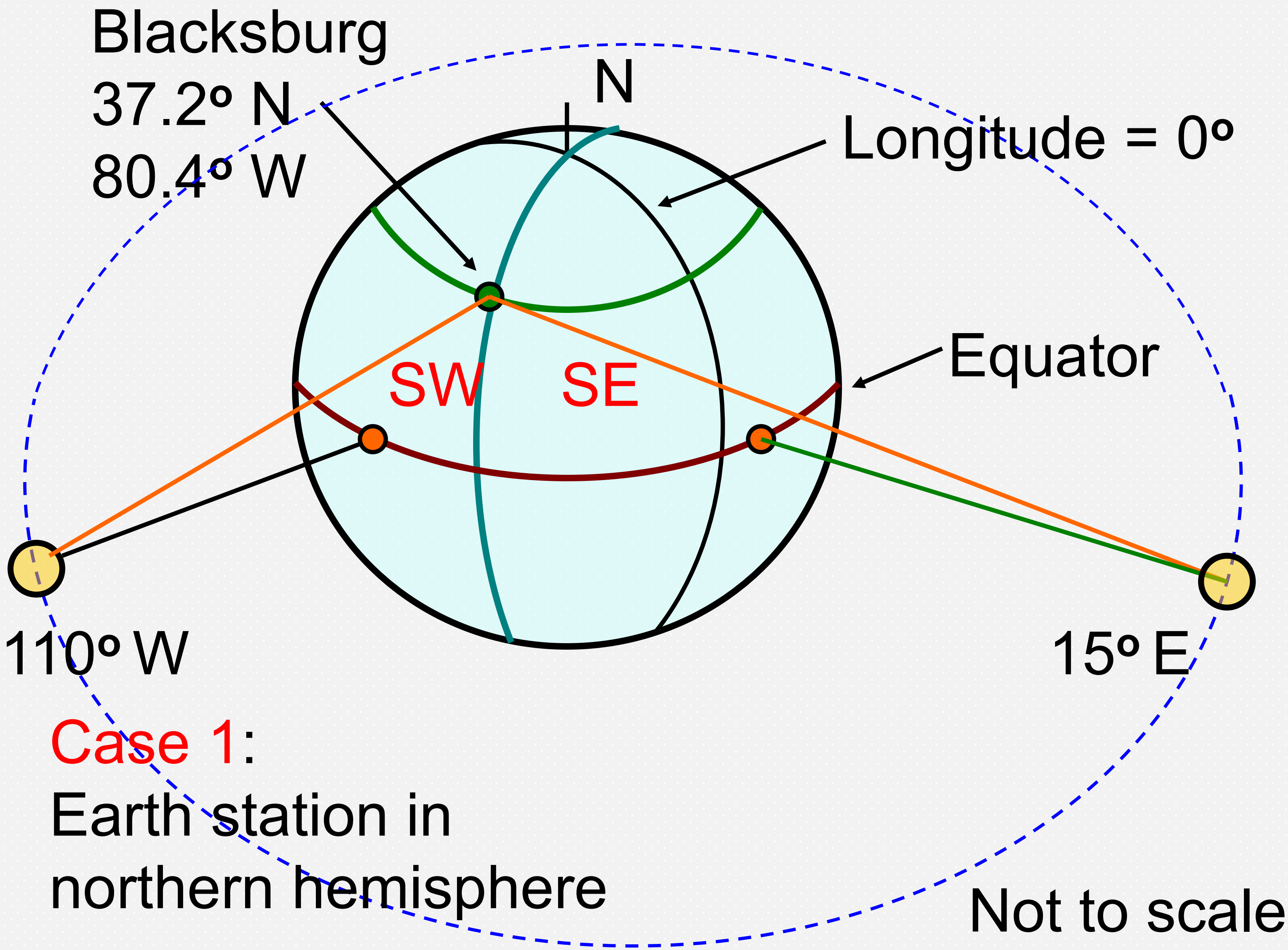
$Az = \alpha$

(d) Satellite to the NW of the earth station:

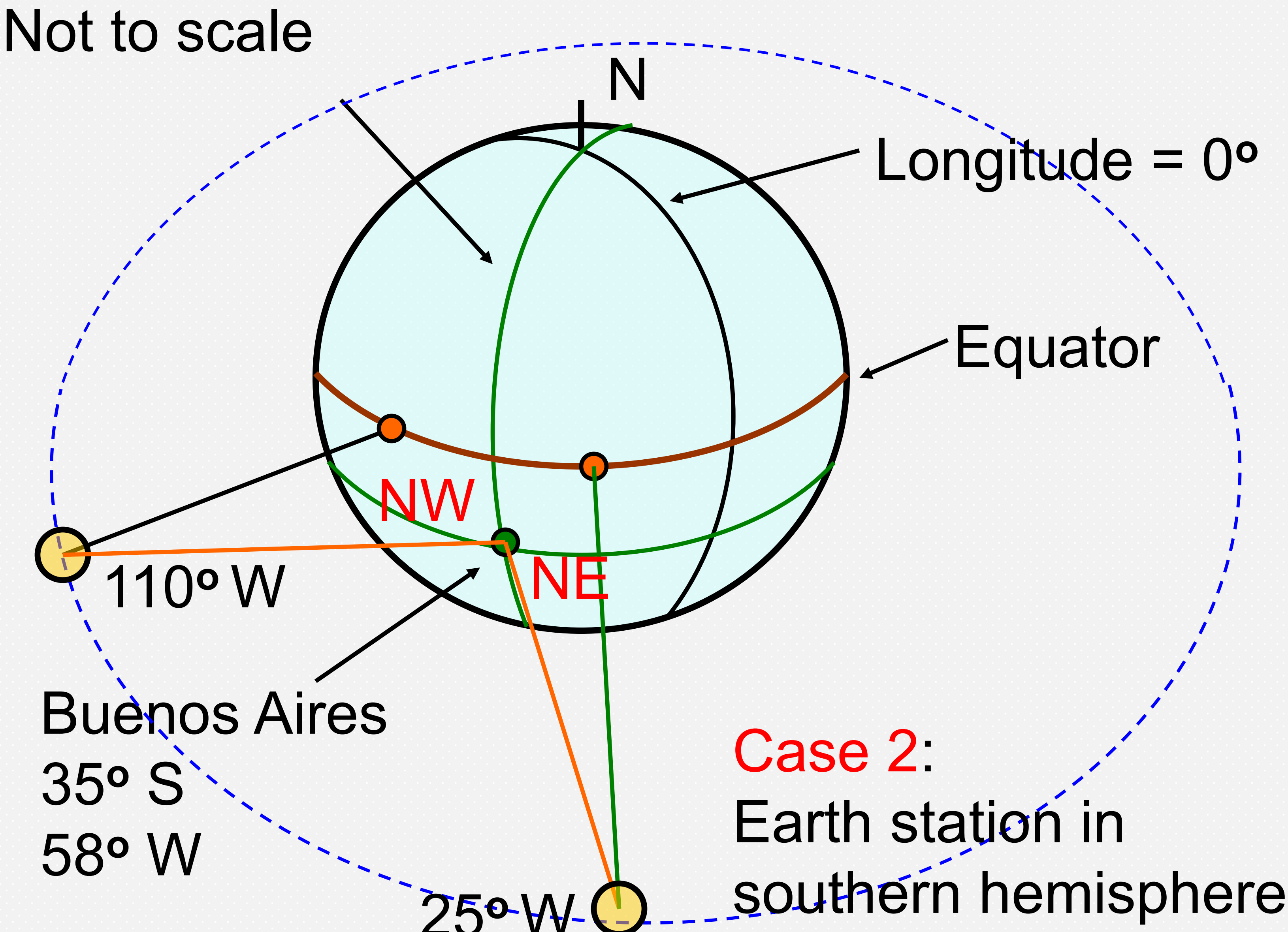
$Az = 360^{o} - \alpha$

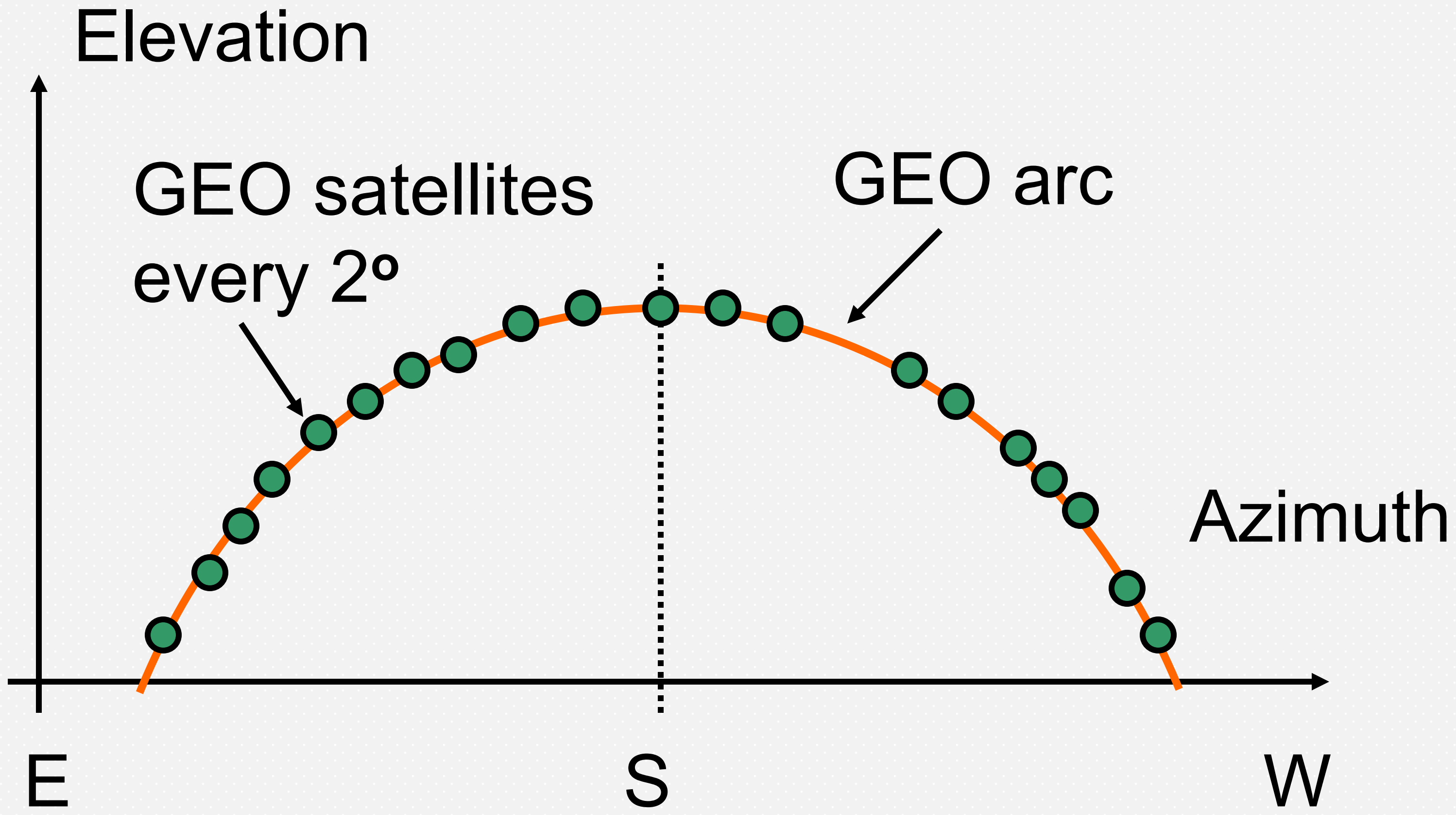
Azimuth Angle Calculation for North American Sector

- Satellite is to SW of earth station when satellite west longitude > earth station west longitude
- Satellite is to SE of earth station when satellite west longitude < earth station west longitude
- Exercise caution when the satellite and earth station are on opposite sides of the prime meridian



Case 1:
Earth station in northern hemisphere


Not to scale
N
Longitude = 0°
Equator
NW
NE
110° W
Buenos Aires
35° S
58° W
25° W
Case 2:
Earth station in
southern hemisphere



GEO arc (Clarke belt) as seen from northern hemisphere. Maximum elevation angle is due south (Az = 180º) where El = 90º – earth station latitude (Blacksburg: El = 52.6º)

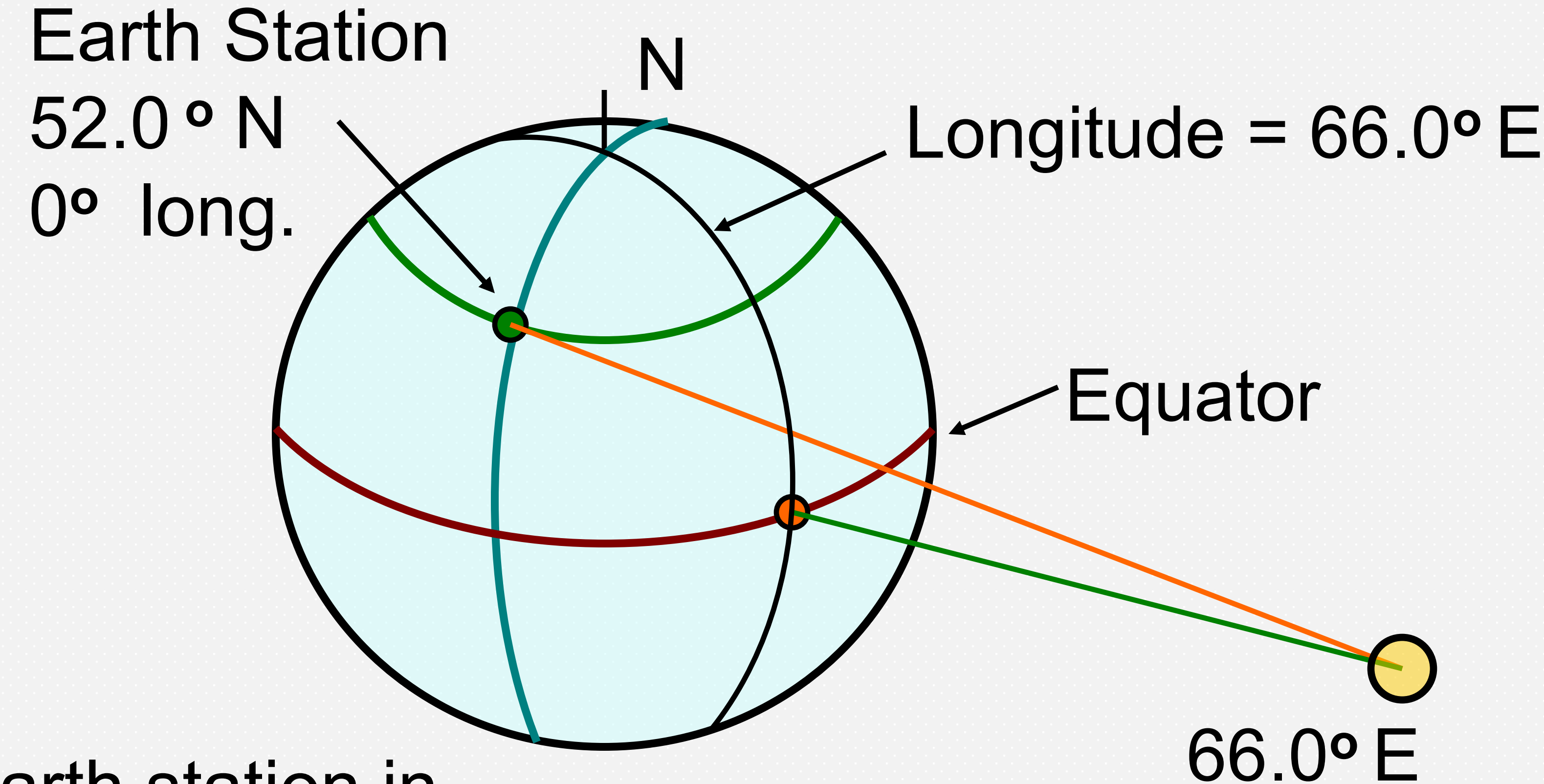
Operational Limitations

- Visibility test
- For geostationary satellites, $\gamma < 81.3^{o}$ to be visible (El > 0^{o})
- Path length through atmosphere is too long at very low elevation angles
- Usual limits are:

C-Band	5^{o}
Ku-Band	10^{o}
Ka- and V-Band	20^{o}

Look Angle Example

- Example 2.2.1
  - Satellite located at 66.0° E (Indian Ocean)
  - Earth station in London, England at 52° N latitude, 0° longitude
- Find look angles



Earth station in northern hemisphere

Satellite is SE of earth station

Not to scale

Look Angle Example

- **Step 1**. Find the central angle γ

$$\cos \gamma = \cos (L_e) \cos (l_s - l_e) \qquad \text{Eqn 2.36}$$

$$= \cos 52^{\circ} \text{ x } \cos 66^{\circ}$$

$$\gamma = 75.4981^{\circ}$$

- Central angle is < 81.3º so satellite is visible
- Large γ indicates a low elevation.

Look Angle Example – 2.2.1

- **Step 2**. Find the elevation angle El

  El = $\tan^{-1}$ [(6.6107345 - cos γ) / sin γ] - γ

  (Eqn 2.39)

  El = $\tan^{-1}$[(6.6107345 - cos75.4981) / sin75.4981]

  - 75.4981º

  El = 81.345º - 75.498º = 5.847º

- As expected, the satellite is not far above the horizor

Look Angle Example – 2.2.1

- **Step 3**. Find the intermediate angle α

  $\alpha = \tan^{-1} [(\tan |(l_s - l_e)|) / \sin L_e]$ (Eqn 2.40)

- The longitude difference angle between the satellite and the earth station is

  $(l_s - l_e) = (66^o \text{ E} – 0^o) = 66^o$

- The latitude of the earth station is $L_e = 52.0^o$

  $\alpha = \tan^{-1} [(\tan 66^o / \sin 52^o] = 70.667^o$

Look Angle Example – 2.2.1

- **Step 4**. Find the azimuth angle
- Intermediate angle $\alpha = 70.667^{\circ}$
- The earth station is in the northern hemisphere (Case 1)
- The satellite is east of the earth station
- Hence the satellite is SE of earth station

$$Az = 180^{\circ} - \alpha = 109.333^{\circ}$$

Look Angle Example – 2.2.1

- Summary of results for look angles
- Satellite is SE of earth station at a low elevation angle

$$El = 5.847^{\circ} \quad \text{(C-band only, marginal)}$$

$$Az = 109.333^{\circ}$$

- Azimuth can also be quoted as 70.667º east of south

Summary (1/2)

- Kepler's three laws describe satellite orbits
- Six orbital elements are required to specify a particular orbit
- Three elements, (a, e, t_p), are needed to solve for satellite position in the orbital plane
- To relate satellite position to an Earth observer, we need the other three elements, (Ω, ω, i)
- Position in the sky is stated in terms of Right Ascension (RA) and declination (δ)

Summary (2/2)

- An Earth observer requires look angles to the satellite
- Calculations will render these as azimuth and elevation
- Straightforward for GEO satellites
- Complicated for LEO and MEO satellites, consult commercial software packages
- Central angle γ must be > 81.3^o for visibility